

Review of last week's lecture

PTGS:

Rdrp:

RISC:

VIGS:

RNAi :

SiRNA :

miRNA :

LncRNA :

Translation

- Translation: the process by which ribosomes read the genetic message in mRNA and produce a protein product according to the messages's instruction.

Components in Translation:

- **Messenger RNA (mRNA)**
- **Ribosomes** (complexes of protein and ribosomal RNA [rRNA])
protein factories
- **Transfer RNA (tRNA)**
- **Various protein factors** : In eukaryotes, >70 ribosomal proteins; >12 enzymes; ≥ 100 proteins for processing; ≈ 50 tRNA and rRNAs (minimum); Other specific proteins
~300 molecules involved
- **Amino acids**
- **Energy (ATP and GTP)**

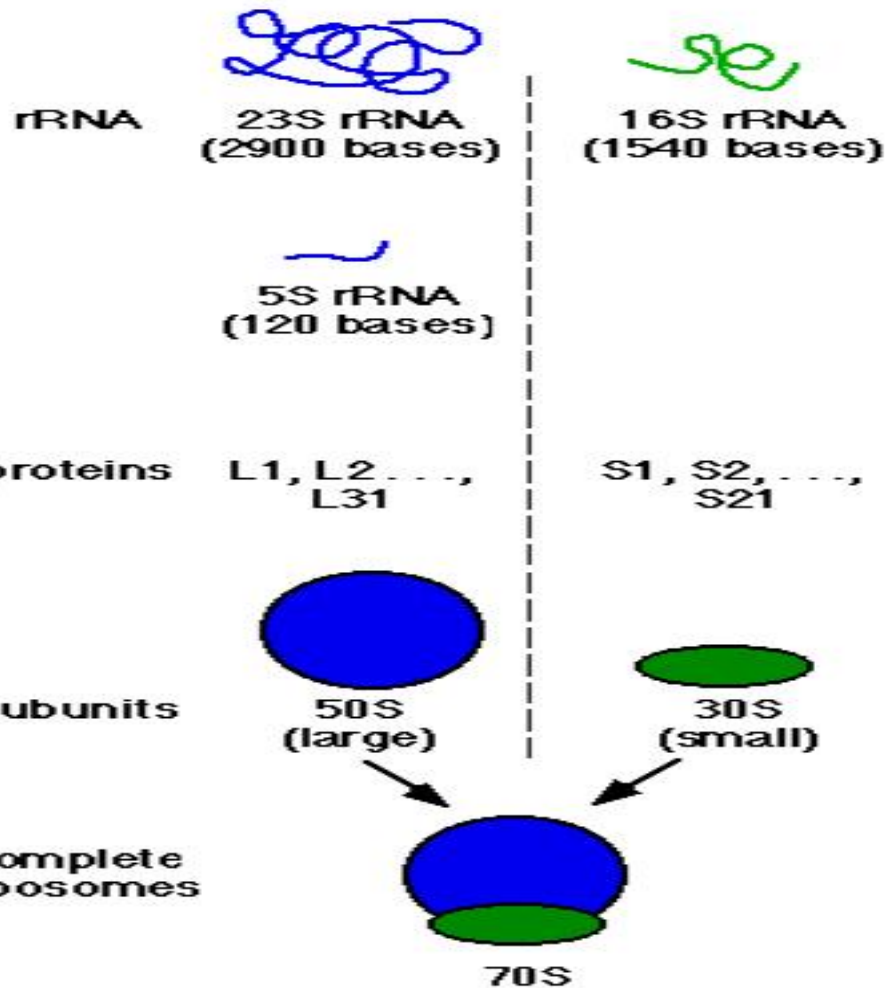
Functions of the different types of RNA

- **mRNA**- serves as a template code
- **tRNA**- serves as an adapter molecule
- **rRNA**- holds molecules in the correct position, protein portion (ribosome) also catalyze reactions

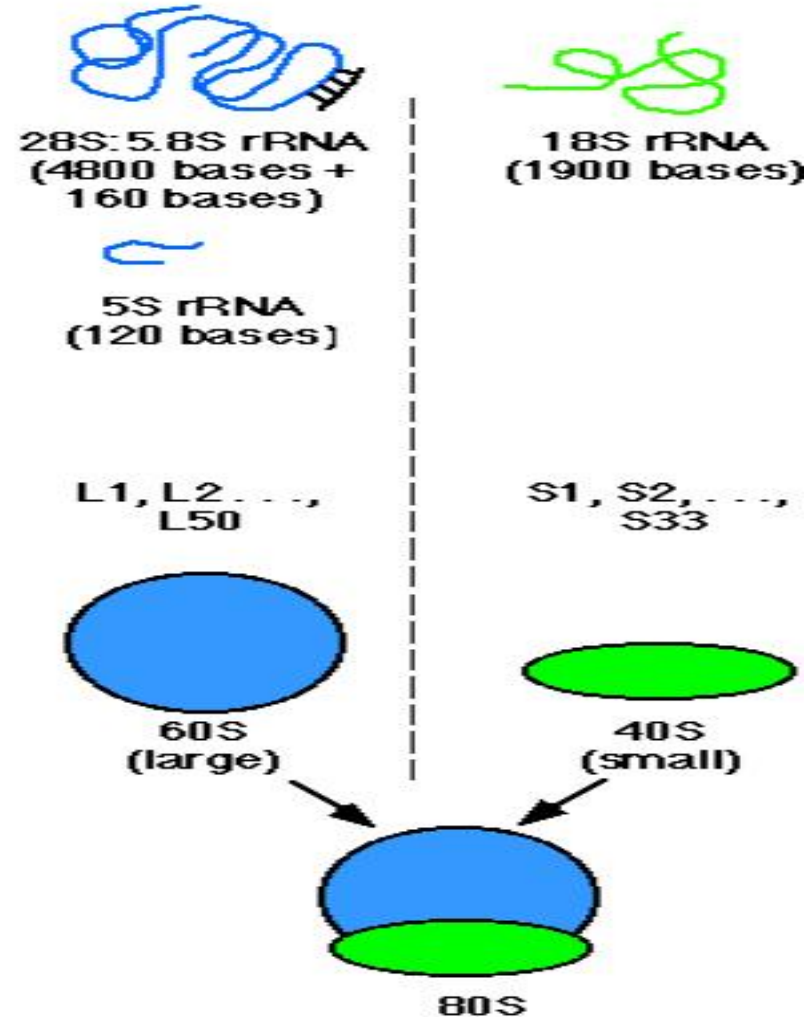
Ribosome Structure

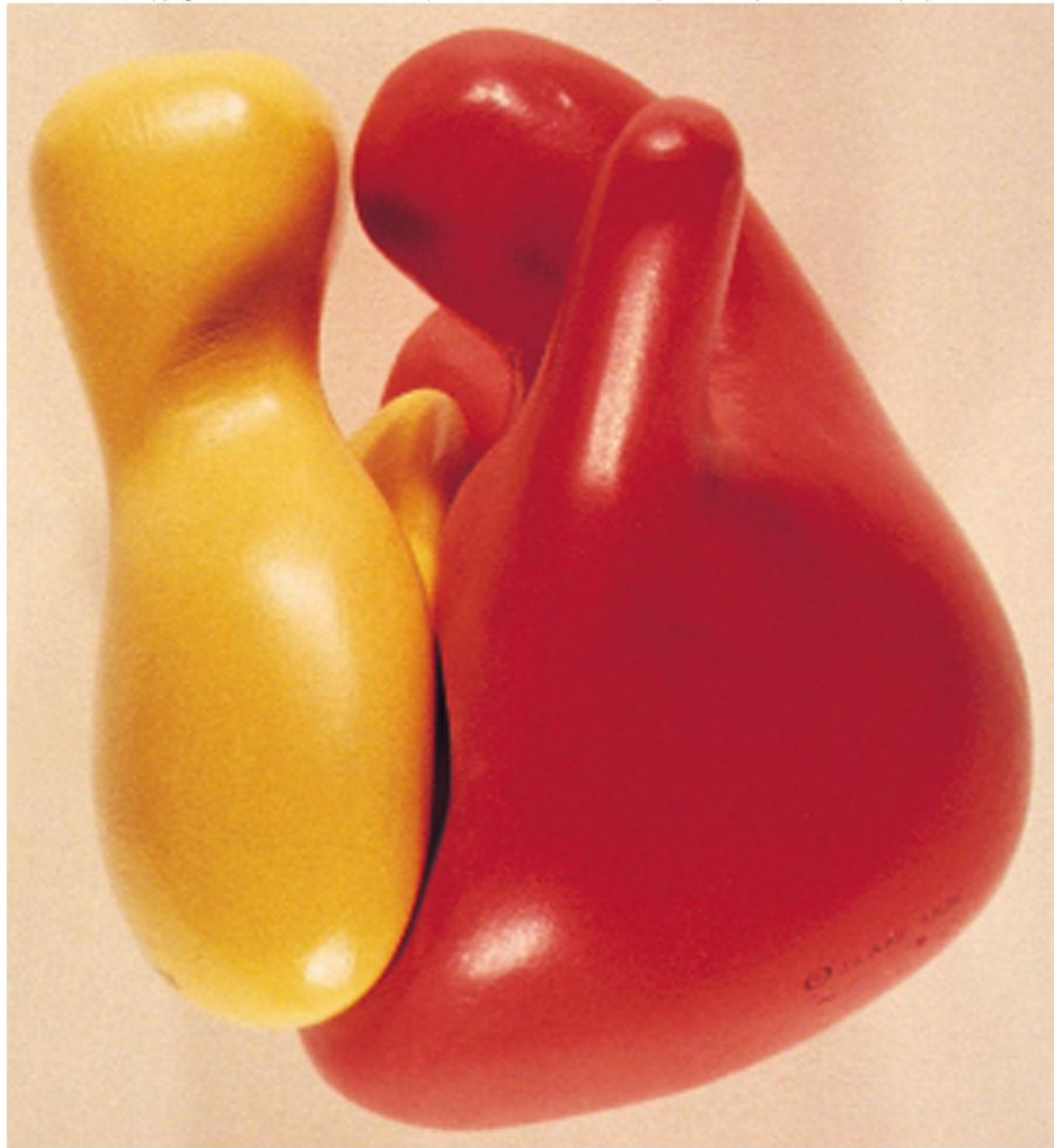
Ribosome Structure

Prokaryotic



Eukaryotic





(Source: Lake, J. Ribosome structure determined by electron microscopy of *Escherichia coli* small subunits, large subunits and monomeric ribosomes. *J. Mol. Biol.* 105 (1976) p. 155, fig. 14, by permission of Academic Press.)

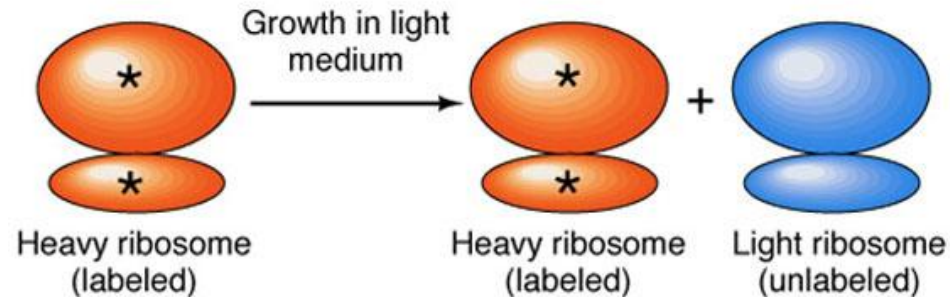
E.Coli ribosome structure

(a)The 70S ribosome is shown from the “side “ of the 30S Particle (yellow) and the 50S particle (red)

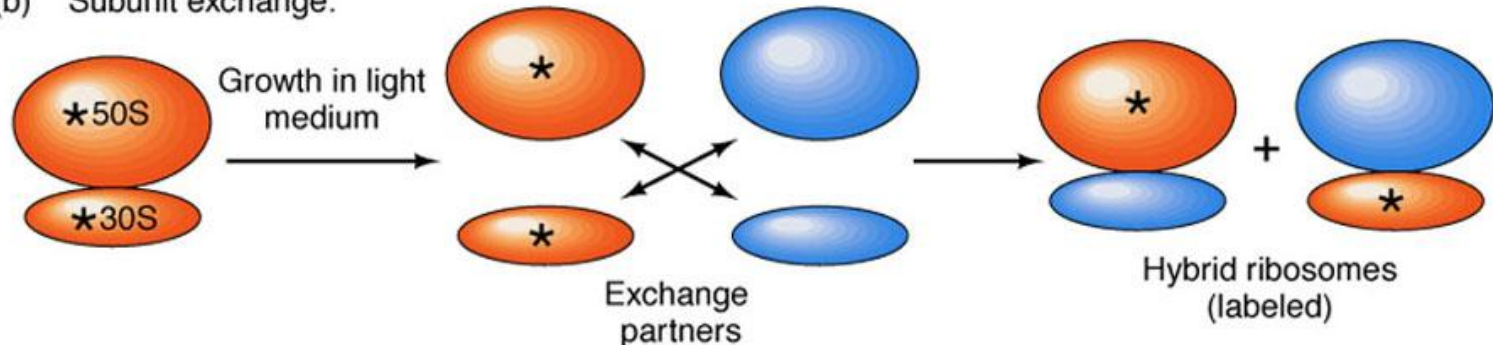
Dissociation of ribosomes

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(a) No exchange:

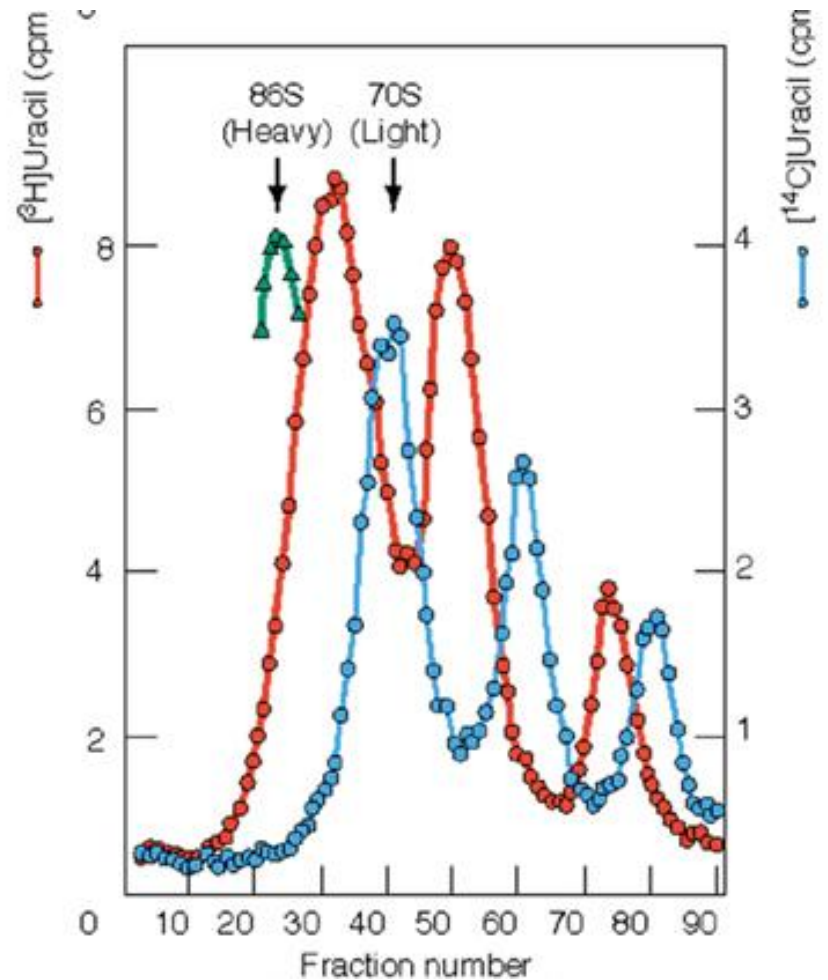
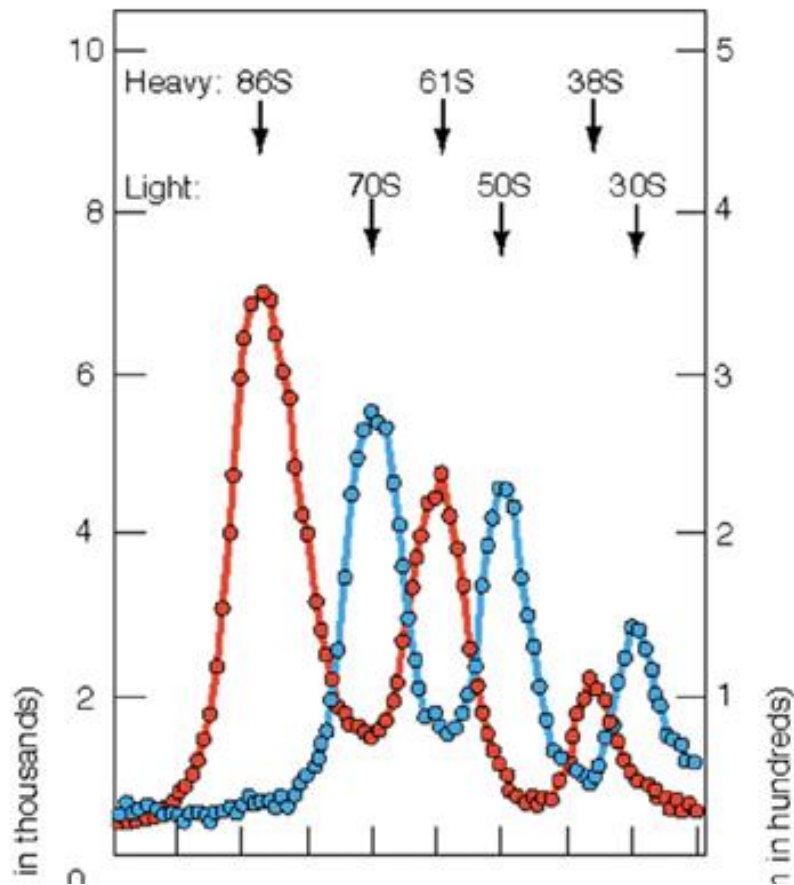


(b) Subunit exchange:

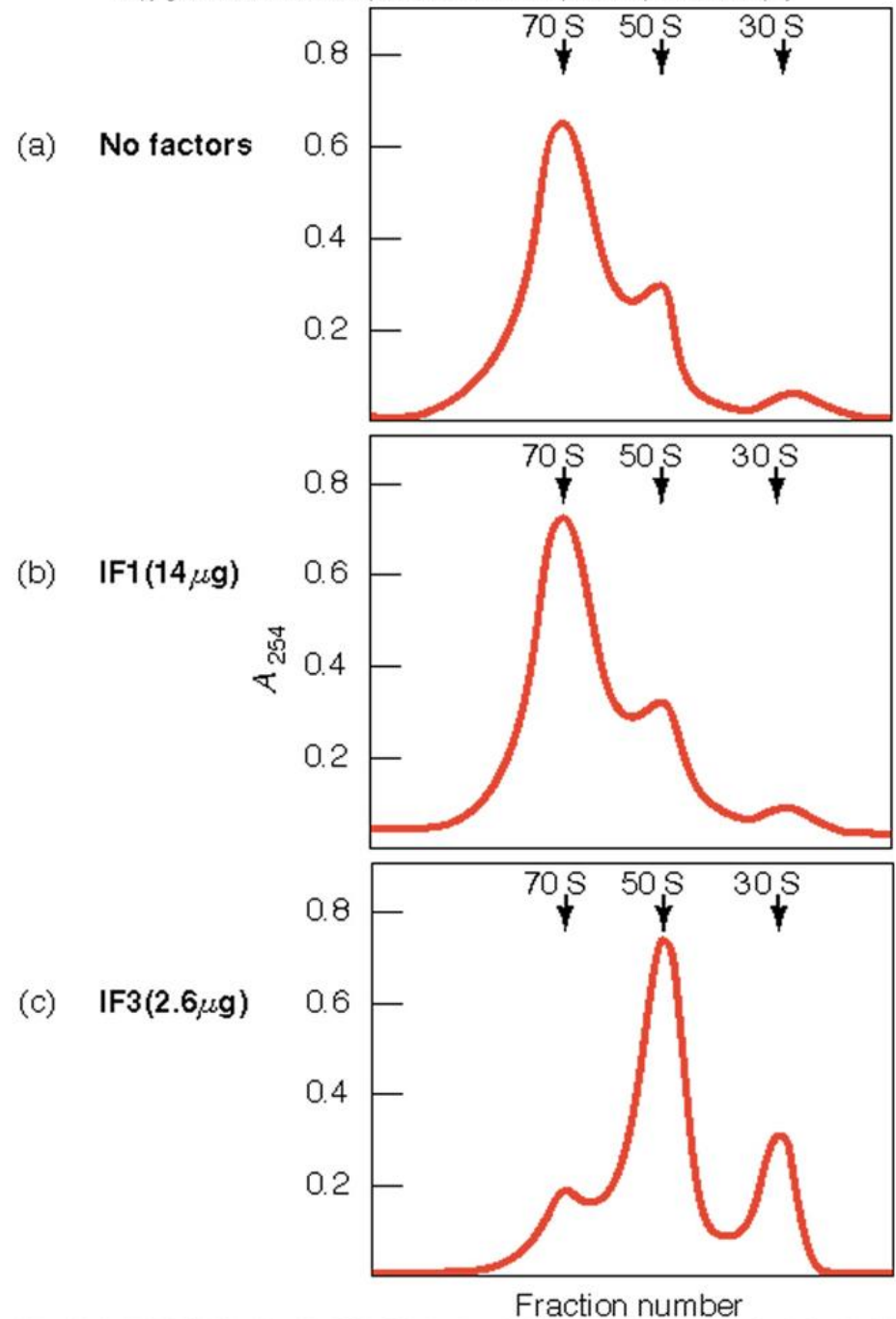


Experimental plan to demonstrate ribosomal subunit exchange

Demonstration of ribosomal subunit exchange

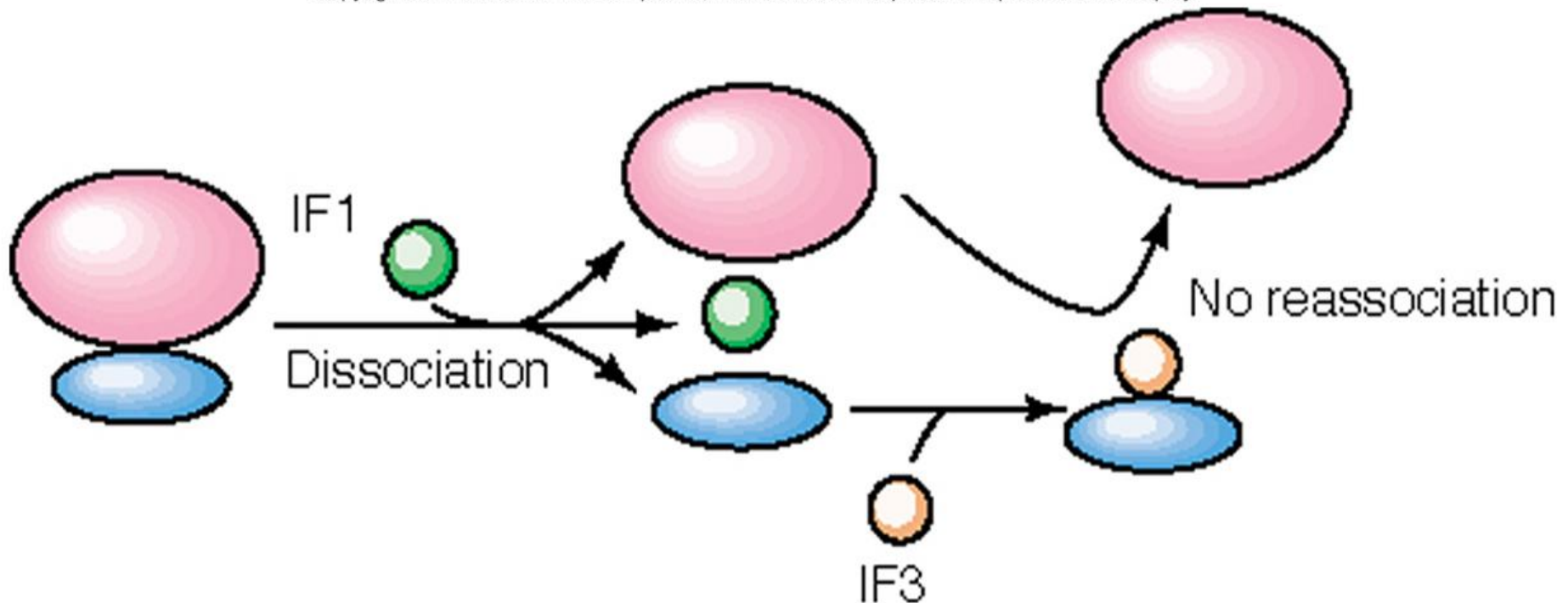


Effect of IF3 in dissociating ribosomes



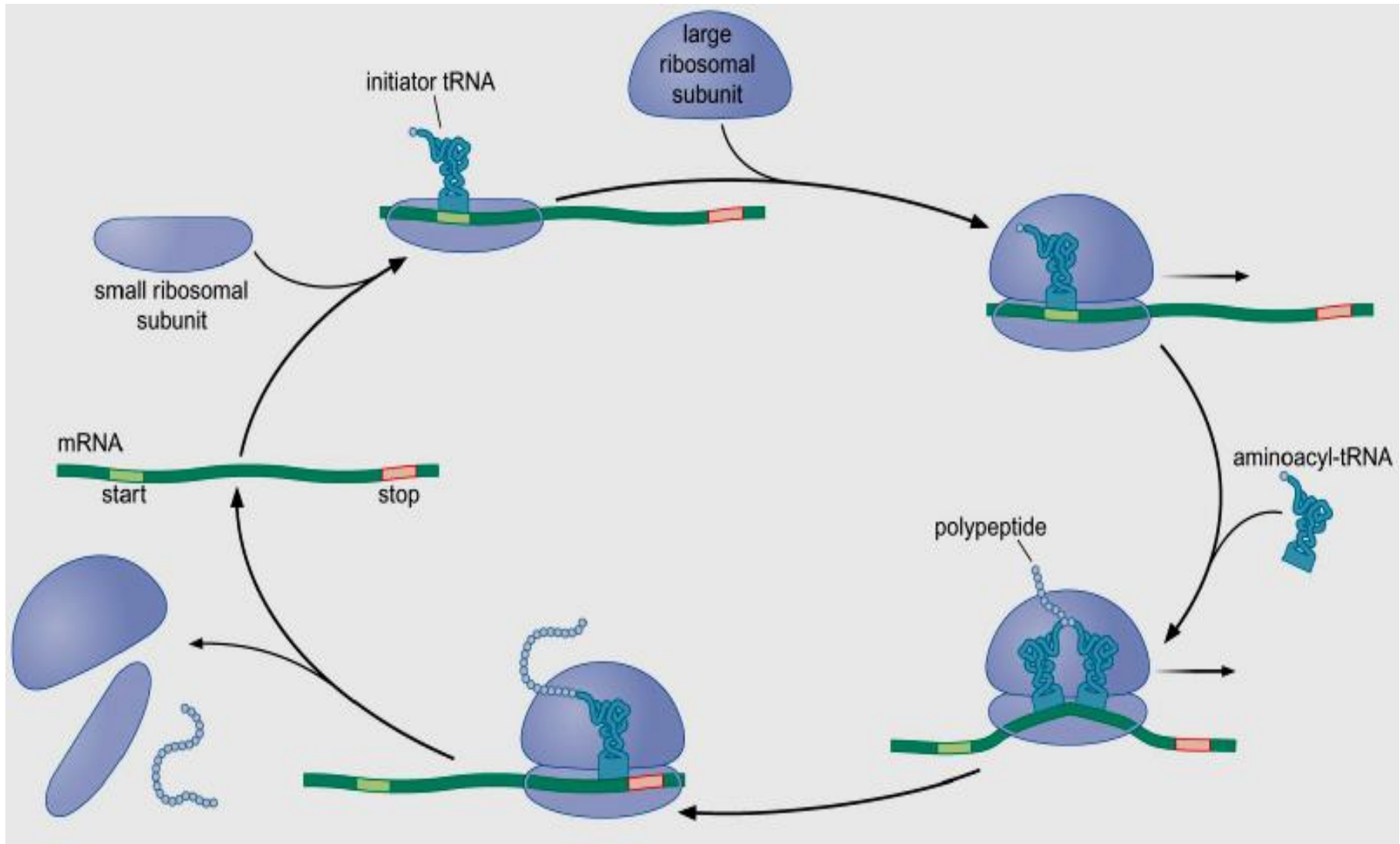
- *E. Coli* ribosomes dissociate into subunits at the end of each round of translation.
- IF1 actively promotes this dissociation, and IF3 binds to the free 30S subunit and prevents its reassociation with a 50S subunit to form a whole ribosome

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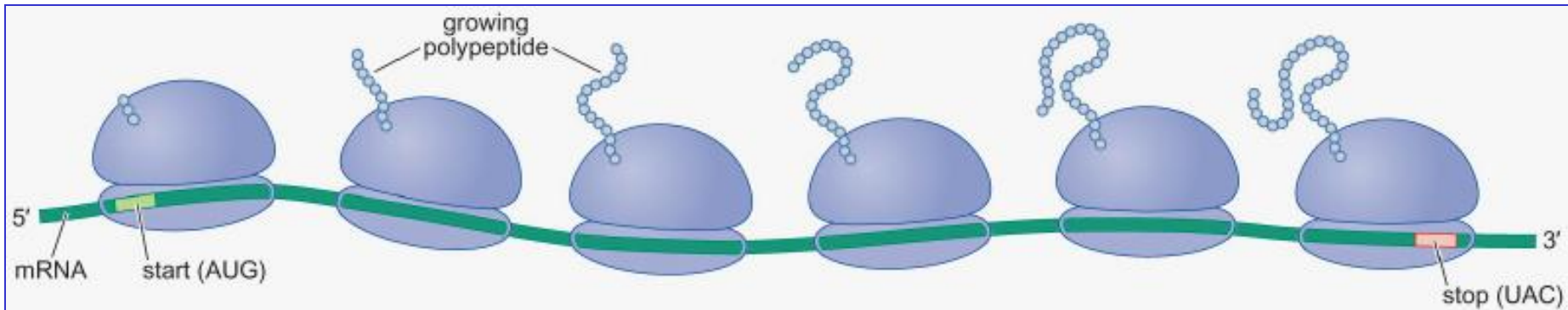


The large and the small subunits undergone association and dissociation during each cycle of translation-

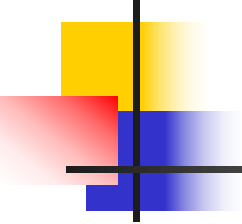
Ribosome cycles



Polysome/polyribosome: an mRNA bearing multiple ribosomes



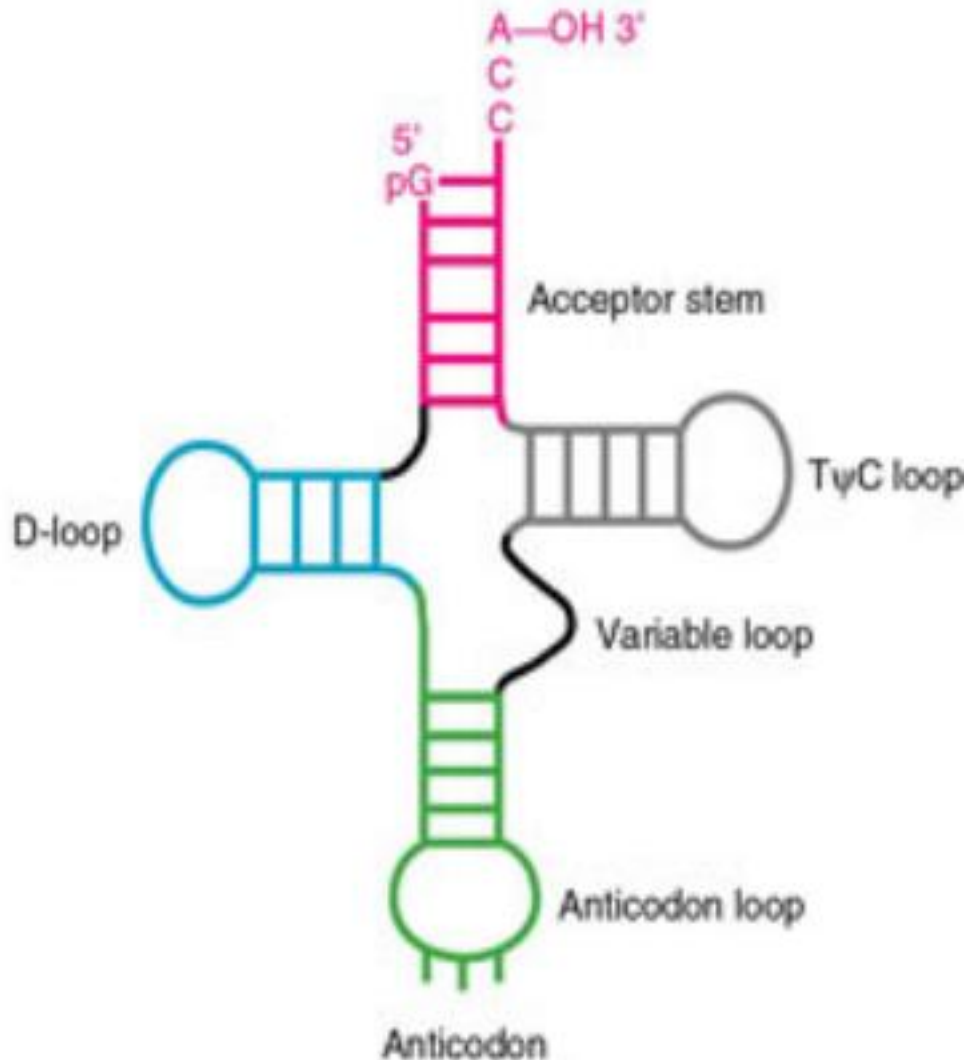
Each mRNA can be translated simultaneously by multiple ribosomes



Main challenge of translation

- The genetic information in mRNA cannot be recognized by amino acids.
- The genetic code has to be recognized by **an adaptor molecular (translator)**, and this adaptor has to accurately recruit the corresponding amino acid.

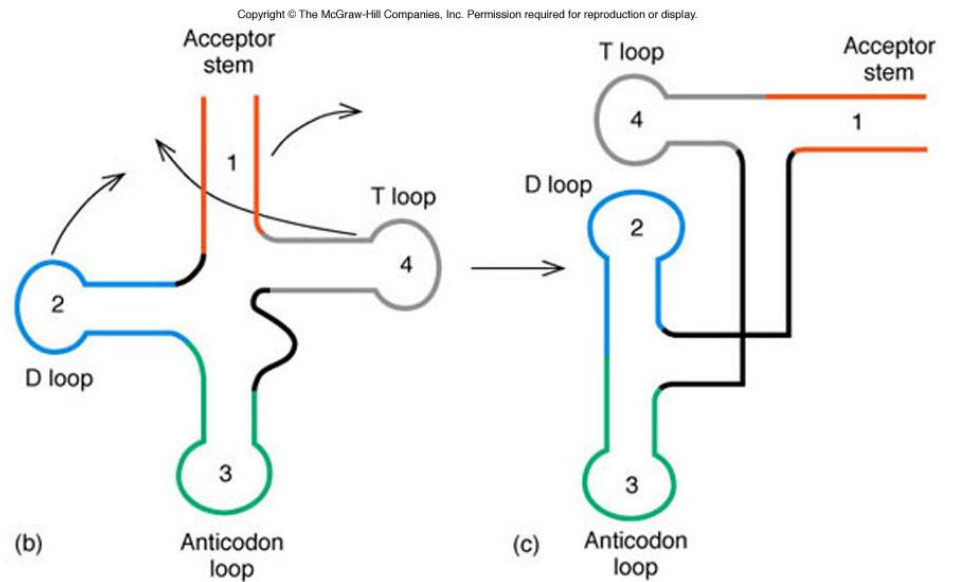
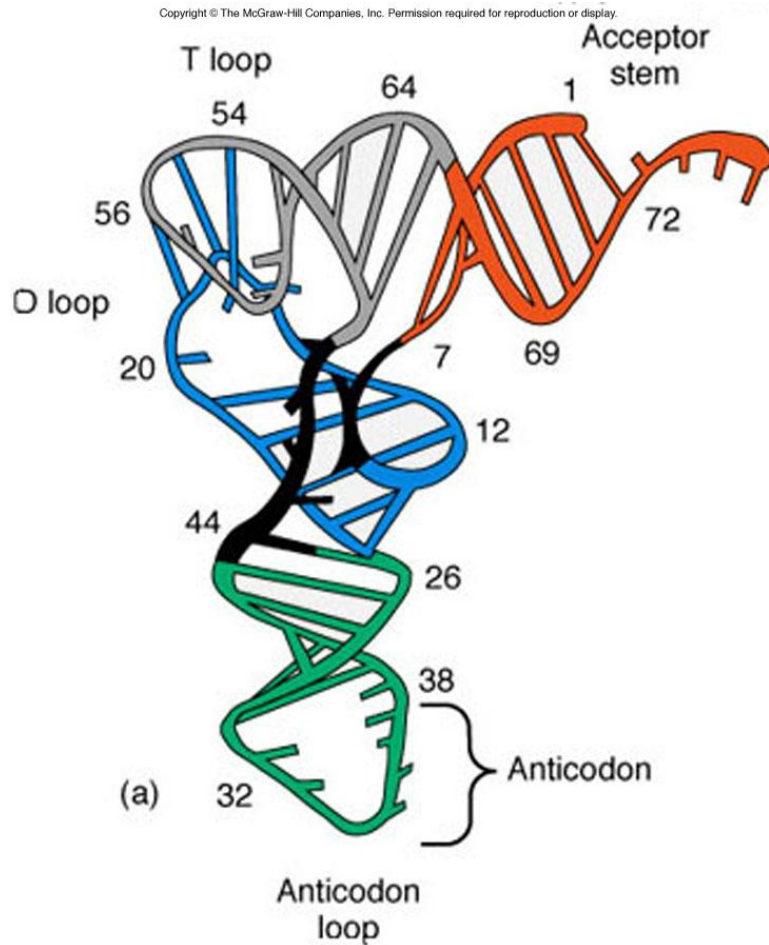
tRNA structure: cloverleaf



- ~75 nt
- Acceptor arm- CCA-3'
- T ψ C arm
- Variable length extra arm
~3-~20 nt
- anti-codon arm
- D arm

D, ψ are modified bases and nucleotides in tRNA

Three-dimensional structure of tRNA^{Phe}.



Translation: five steps

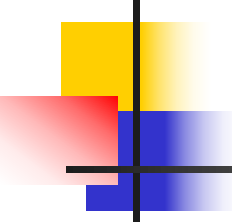
- Preinitiation: tRNA charging (adding the correct amino acid to the correct tRNA); Dissociation of ribosomes
- Initiation:(assembly mRNA, ribosomes, and tRNA)
- Elongation:(reading the code and converting the information into a peptide)
- Termination: ending the synthesis of a protein
- Post-translational modification

Initiation of translation: Prokaryotes

Association of mRNA, tRNA and ribosomes

Two important events must occur before translation initiation can take place

- Generate a supply of aminoacyl-tRNAs
 - Amino acids must be covalently bound to tRNAs
 - Process of bonding tRNA to amino acid is called **tRNA charging**
- Dissociation of ribosomes into their two subunits
 - The cell assembles the initiation complex on the small ribosomal subunit
 - The two subunits must separate to make assembly possible



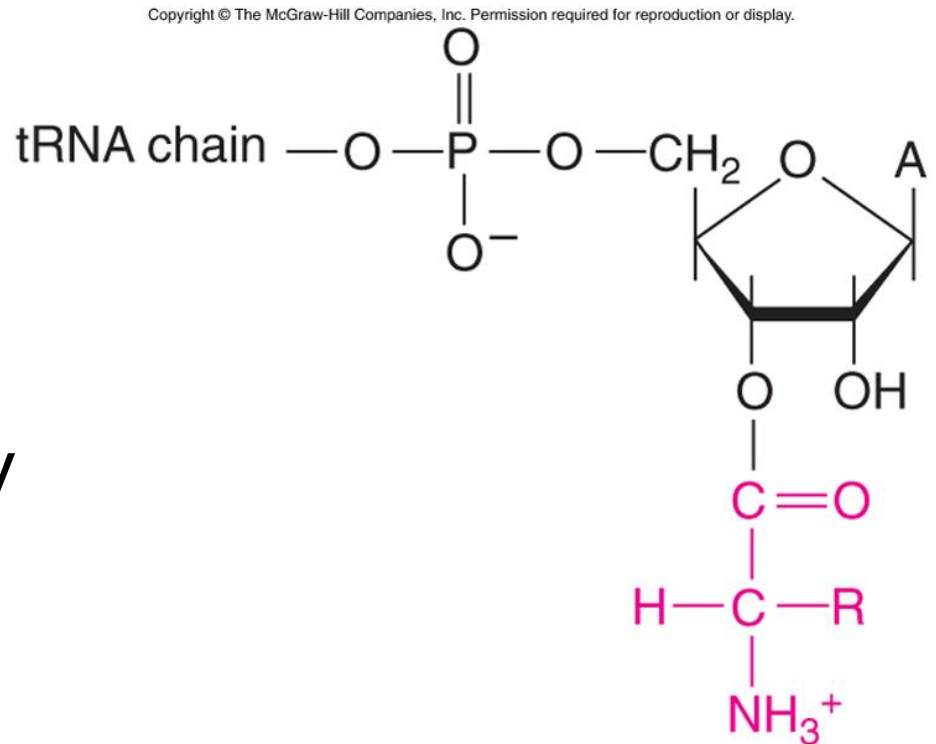
tRNA Charging

- tRNA charging: the process of adding the correct amino acid to the tRNA.
- The fidelity of protein synthesis heavily rely on the accuracy of this charging.
- Aminoacyl tRNA synthetases catalyze the process

amino acid (aa) + ATP + tRNA --> aminoacyl-tRNA + AMP + 2 Pi

tRNA Charging

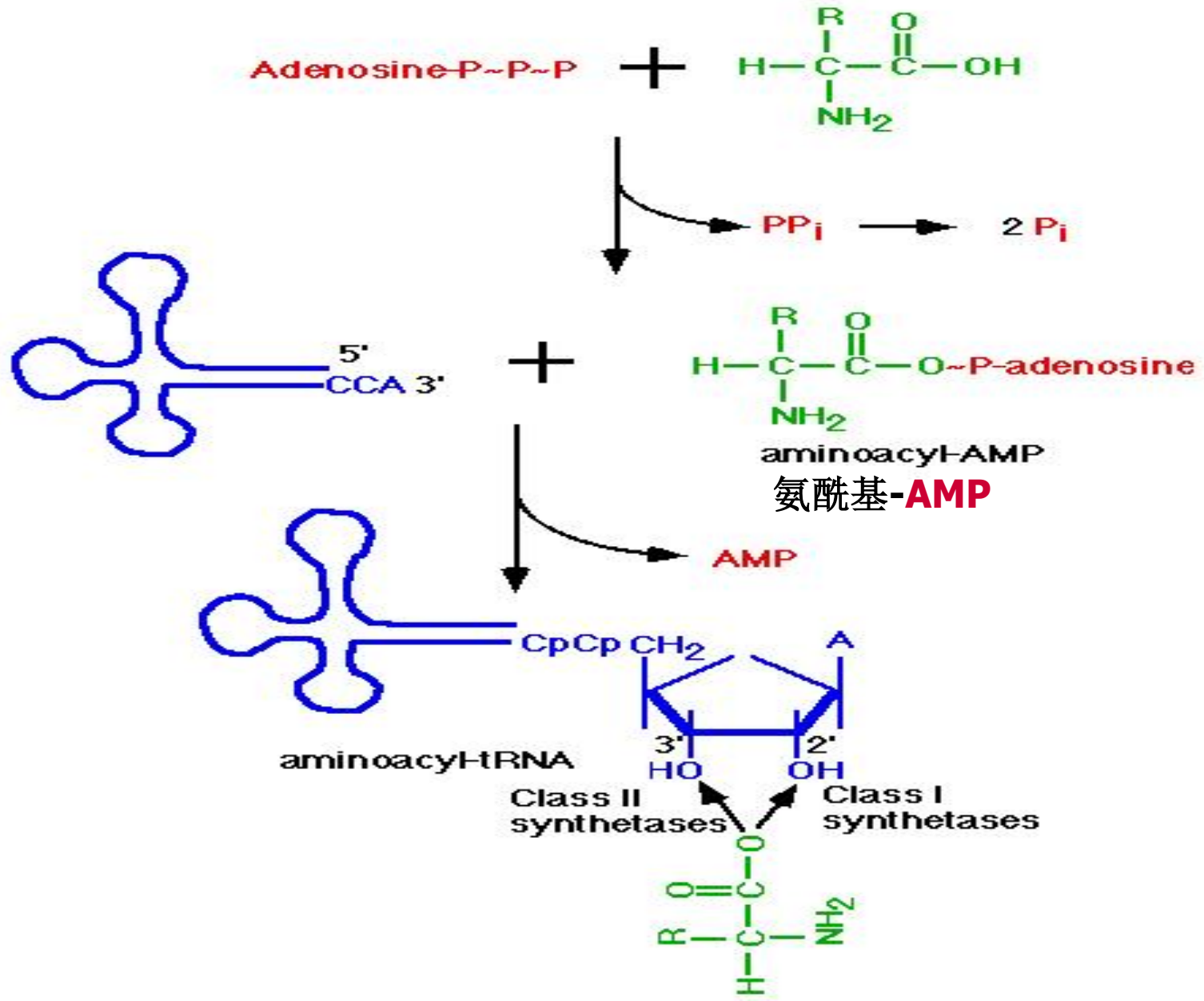
- All tRNAs have same 3 bases at 3'-end (CCA)
- Terminal adenosine is the target for charging with amino acid
- Amino acid attached by ester bond between
 - Its carboxyl group
 - 2'- or 3'-hydroxyl group of terminal adenosine of tRNA

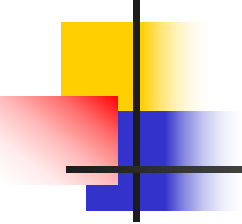


Two-Step Charging

- Begins with activation of the amino acid with AMP derived from ATP
- In the second step, the energy from the aminoacyl-AMP is used to transfer the amino acid to the tRNA

Reaction Catalyzed by Aminoacyl-tRNA Synthetases





Dissociation of Ribosomes

- Ribosomes dissociate into subunits at the end of each round of translation
- RF and EF-G actively promotes this dissociation
- IF3 binds to free 30S subunit and prevents reassociation with 50S subunit to form a whole ribosome

Formation of the 30S Initiation Complex

Once the ribosomal subunits have been dissociated, the cell builds a complex on the 30S subunit:

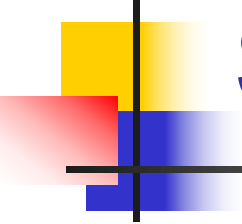
- mRNA
- Aminoacyl-tRNA
- Initiation factors
 - IF3 binds by itself to 30S subunit
 - IF1 and IF2 stabilize this binding
 - IF2 can bind alone, but is stabilized with help of IF1 and IF3
 - IF1 does not bind alone

First Codon and the First Aminoacyl-tRNA

- Prokaryotic initiation codon is:
 - Usually AUG
 - Can be GUG
 - Rarely UUG
- Initiating aminoacyl-tRNA is *N*-formyl-methionyl-tRNA
- *N*-formyl-methionine (fMet) is the first amino acid incorporated into a polypeptide
- This amino acid is frequently removed from the protein during maturation

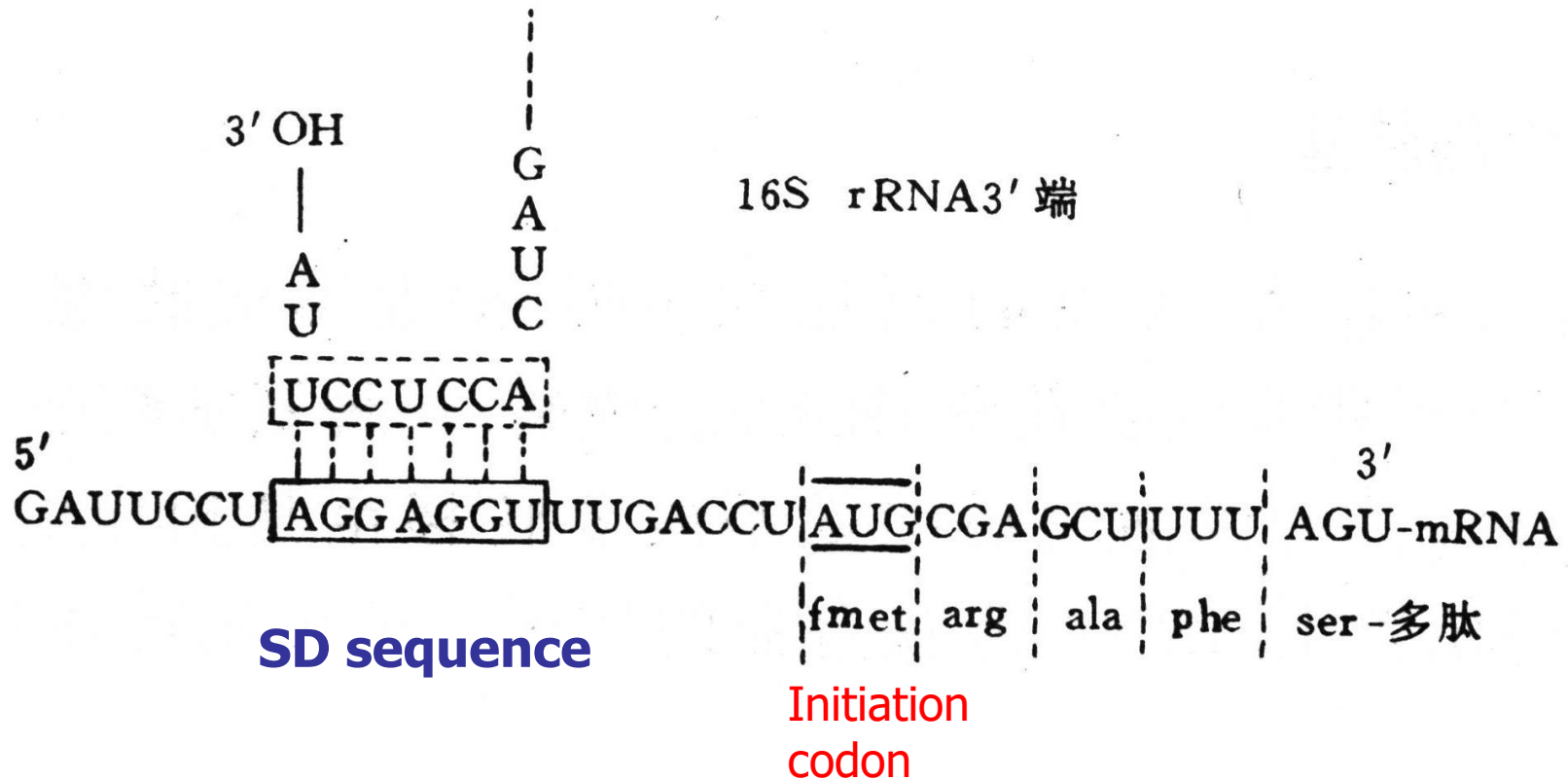
Binding mRNA to the 30S Ribosomal Subunit

- The 30S initiation complex is formed from a free 30S ribosomal subunit plus mRNA and fMet-tRNA
- Binding between the 30S prokaryotic ribosomal subunit and the initiation site of a message depends on base pairing between
 - Short RNA sequence
 - Shine-Dalgarno sequence upstream of initiation codon
 - Complementary sequence
 - 3'-end of 16S RNA

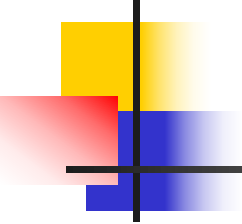


SD sequence

- How does the cell detect the difference between an initiation codon and the same sequence in the interior of a mRNA?
- Shine-Dalgarno (SD) sequence: a G-rich sequence (consensus=AGGAGGU) that is complementary to a sequence at 3'-end of 16S rRNA in *E.coli*
- SD sequence is located ~10 nt upstream of initiation codon

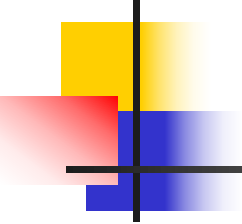


Shine-Dalgarno sequence just upstream of the initiation codon is complementary with 3'-end of the 16S rRNA



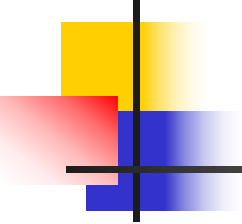
Initiation Factors and 30S Subunit

- Binding of the Shine-Dalgarno sequence with the complementary sequence of the 16S rRNA is mediated by IF3
 - Assisted by IF1 and IF2
 - All 3 initiation factors have bound to the 30S subunit at this time



Binding of fMet-tRNA to the 30S Initiation Complex

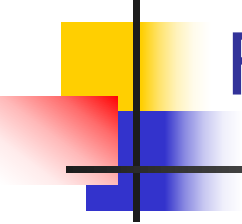
- IF2 is the major factor promoting binding of fMet-tRNA to the 30S initiation complex
- Two other initiation factors also play an important supporting role
- GTP is also required for IF2 binding at physiological IF2 concentrations but GTP is not hydrolyzed in the process



30S Initiation Complex

The complete 30S initiation complex contains one each:

- 30S ribosomal subunit
- mRNA
- fMet-tRNA
- GTP
- Factors IF1, IF2, IF3



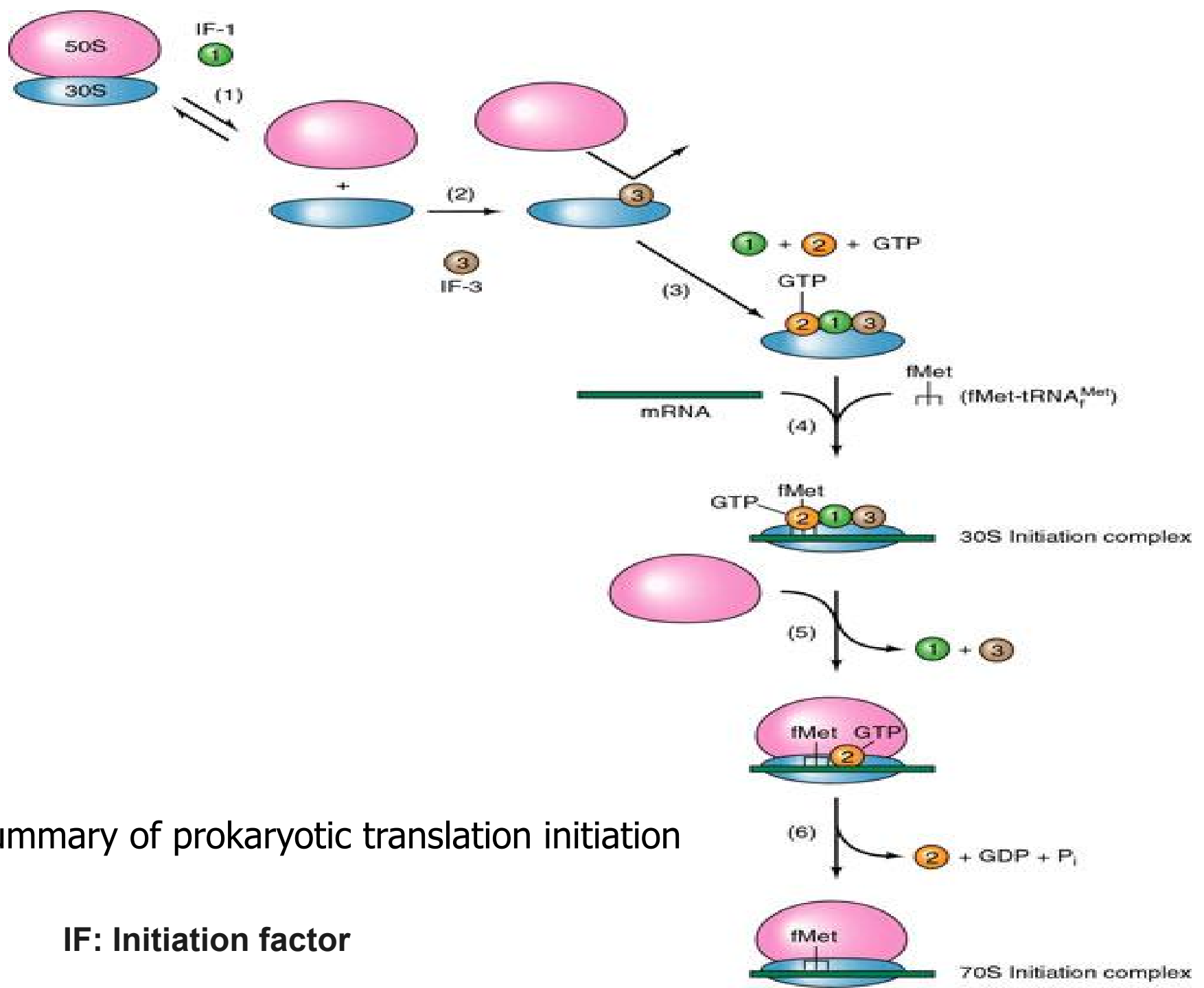
Formation of the 70S Initiation Complex

- GTP is hydrolyzed after the 50S subunit joins the 30S complex to form the 70S initiation complex
- Hydrolysis purpose is to release IF2 and GTP from the complex so polypeptide chain elongation can begin

Initiation of translation: Prokaryotes

Association of mRNA, tRNA and ribosomes

- 1. Dissociation of the 70S ribosome into the 30S and 50S subunits (IF-1)**
- 2. Binding of IF-3 to 30S, prevents reassociation between subunits.**
- 3. Binding of IF-1 and IF-2/GTP to 30S**
- 4. Joining of 30S/IF's with initiator tRNA and mRNA to form 30S initiation complex, (Shine-Dalgarno sequence)**
- 5. Binding of 50S to 30S complex, with loss of IF-1 and IF-3**
- 6. Dissociation of IF-2 with hydrolysis of GTP to GDP and Pi**



Summary of prokaryotic translation initiation

Difference between prokaryotic and eukaryotic translation initiation

■ Eukaryotic

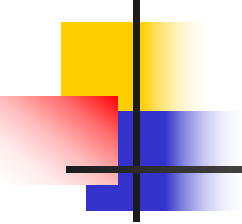
- Begins with methionine
- No Shine-Dalgarno
- mRNA have caps at 5'end

■ Prokaryotic

- N-formyl-methionine
- Shine-Dalgarno sequence to show ribosomes where to start

Scanning Model of Initiation

- Eukaryotic 40S ribosomal subunits locate start codon by binding to 5'-cap and scanning downstream to find the 1st AUG in a favorable context
- Best context uses A of ACCAUGG as +1:
 - Purine in -3 position
 - G in +4 position
- 5-10% of the time, most ribosomal subunits bypass 1st AUG scanning for a more favorable one

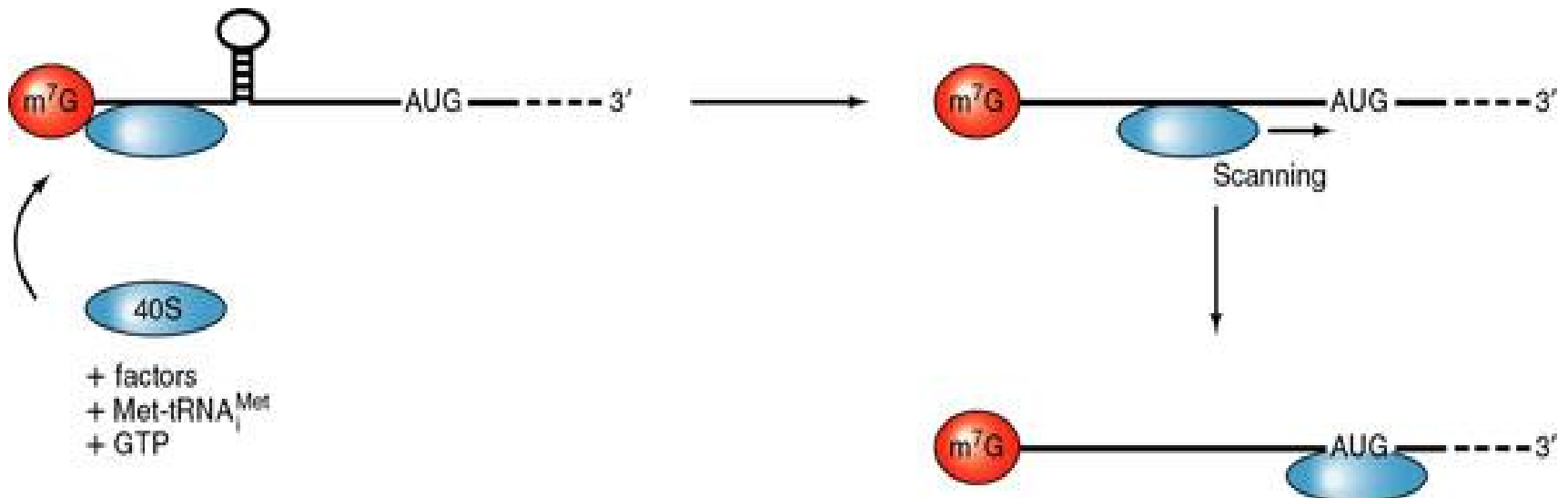


Translation With a Short ORF

- Sometimes ribosomes can use a short upstream open reading frame:
 - Initiate at an upstream AUG
 - Translate a short Open Reading Frame (ORF)
 - Continue scanning
 - Reinitiate at a downstream AUG

Initiation of translation: Eukaryotes

- 40S ribosomal subunit starts at the 5' end and “scans” down the mRNA until the AUG codon
- Consensus sequence of start codon: CCRCCAUGG
R is a purine (A or G)

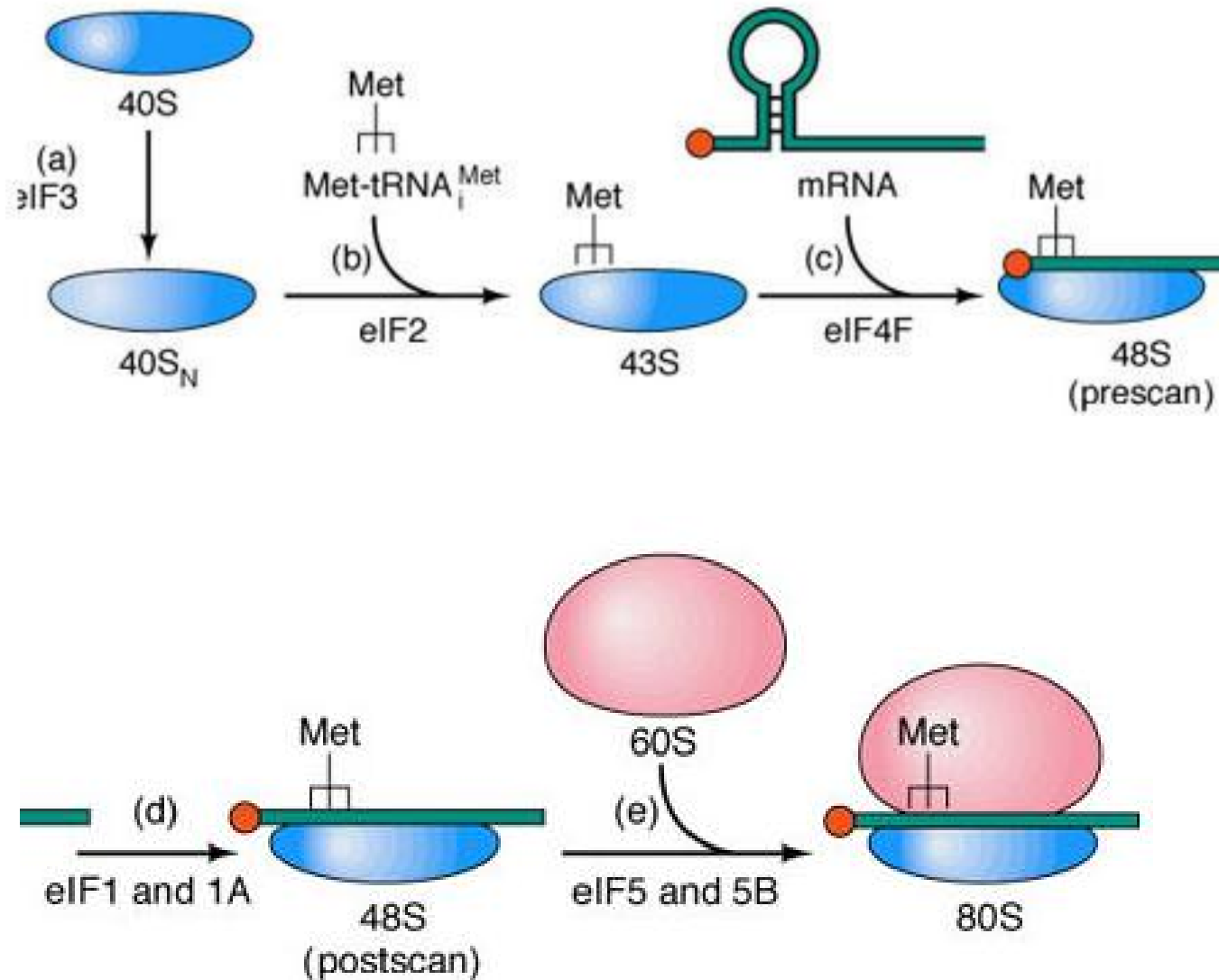


Initiation of translation: eukaryotes

1. 40S subunit joins with eIF-3 to form the 40S_N structure that can join with the initiator tRNA.
2. Along with eIF-2, the tRNA joins to form the 43S complex binding to the capped 5' end of mRNA with the help of eIF-4 to form 48S initiation complex
3. Complex scans down the mRNA until the correct AUG start codon.
4. 60S subunit joins, with the help of eIF-5, to form the 80S initiation complex.

Summary of eukaryotic translation initiation

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Initiation factors (1)

- ♦ **Eukaryotic** factors are designated with the **prefix "e"**.
- ♦ **Initiation** of protein synthesis is much **more complex** in eukaryotes, & requires a large number of protein factors.
- ♦ **Some** eukaryotic **initiation factors** (e.g., **eIF3** & **eIF4G**) serve as **scaffolds**, with multiple domains that bind other proteins during assembly of large initiation complexes.

Initiation factors (2)

- ◆ After the **initiation complex** assembles, it **translocates along** the **mRNA** in a process called **scanning**, until the initiation codon is reached.
- ◆ Scanning is facilitated by eukaryotic initiation factor **eIF4A**, which functions as an **ATP-dependent helicase** to **unwind mRNA** secondary structure while releasing bound proteins.
- ◆ A short sequence of bases adjacent to the AUG initiation codon may aid in **recognition** of the start site.
CCRCCAUGG R is a purine (A or G)
- ◆ After the initiation codon is recognized, there is **hydrolysis of GTP** and **release of initiation factors**, as the **large ribosomal subunit** joins the complex and elongation commences.

Summary

- The eukaryotic initiation factors have the following general functions:
- eIF2 is involved in binding Met-tRNA_i^{Met} to the ribosome. eIF2B activates eIF2 by replacing its GDP with GTP.
- eIF1 and eIF1A aid in scanning to the initiation codon
- eIF3 binds to the 40S ribosomal subunit inhibits its reassociation with 60s.
- eIF4F allows the 40S to bind to the 5'-end of an mRNA.
- eIF5 promotes association between the 60S and the 48S complex.
- eIF6 binds to 60s and blocks its reassociation with 40s

Elongation

1. In what direction does the ribosome read the mRNA? In what direction is a polypeptide synthesized?

5' → 3'; amino → carboxyl

2. What is the nature of the genetic code that dictates which aa will be incorporated in response to the mRNA?

Genetic Codons

Genetic code: the correspondence between the sequence of the four bases in nuclei acids and the sequence of the 20 amino acids in protein.

The set of three-base code words (codons) in mRNAs that stand for the 20 amino acids in proteins.

Second position

First position (5'-end)

	U	C	A	G	
U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA } STOP UAG } STOP	UGU } Cys UGC } UGA } STOP UGG } Trp	U C A G
C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	U C A G
A	AUU } AUC } Ile AUA } AUG } Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G
G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U C A G

Third position (3'-end)

Genetic Codons

- Triplet : A three base sequence (codon) specifies an aa.
- Commaless: There are no untranslated gaps in the code.
- Unambiguous: A particular codon specifies only one aa
- Degenerate: More than one triplet codes for a given amino acid.
- Start codon: AUG=Met
- Stop codon: UAA, UAG, UGA

Codon preference

- The idea that for amino acids with several codons. One or a few of the codons are preferred and are used disproportionately.
- They would correspond with tRNAs that are abundant.

ORF: Open Reading Frame

- **ORF: the sequence of codons from (and including) the start codon (usually AUG) to the stop codon (UAG, UAA, or UGA) to code for legitimate protein.**
- **a sequence could be read in 3 ORFs**

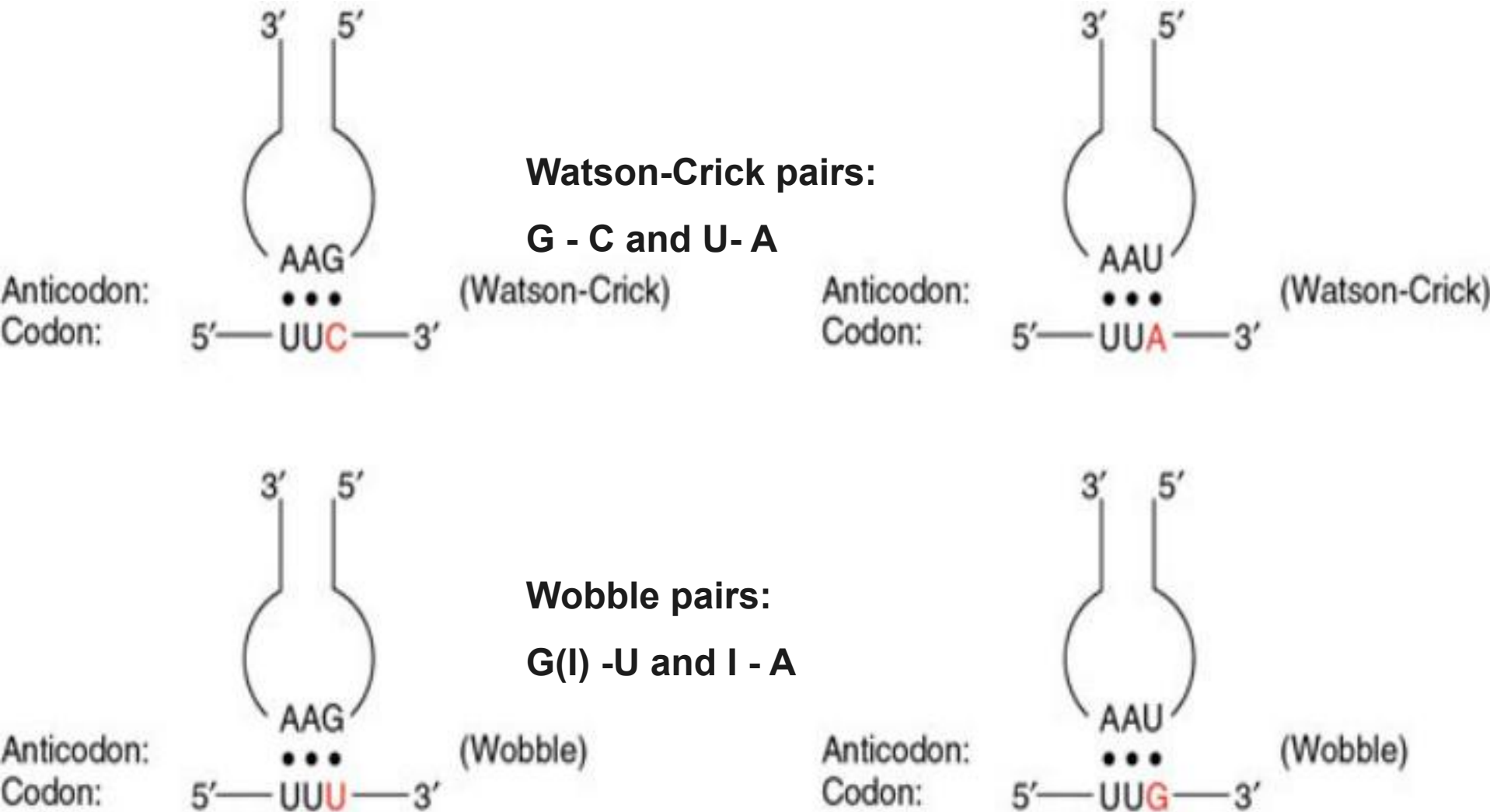
ACUGUCGUC...

1. ACU GUC GCU C...
2. CUG UCG CUC...
3. UGU CGC UC...
4. GUC GCU C...

Wobble hypothesis (摆动假说)

- In 1966, Francis Crick hypothesized that the first two bases of a codon must pair correctly with the anticodon according to Watson-Crick base-pairing rules, but the last base of codon can wobble from its normal position to form unusual base pair with the anticodon.
- **Wobble base pair:** The ability of the third base of a codon to shift slightly to form a non-Watson-Crick base pair with the first base of an anticodon, thus allowing a tRNA to translate more than one codon.

Base Pairs between codon and anticodon



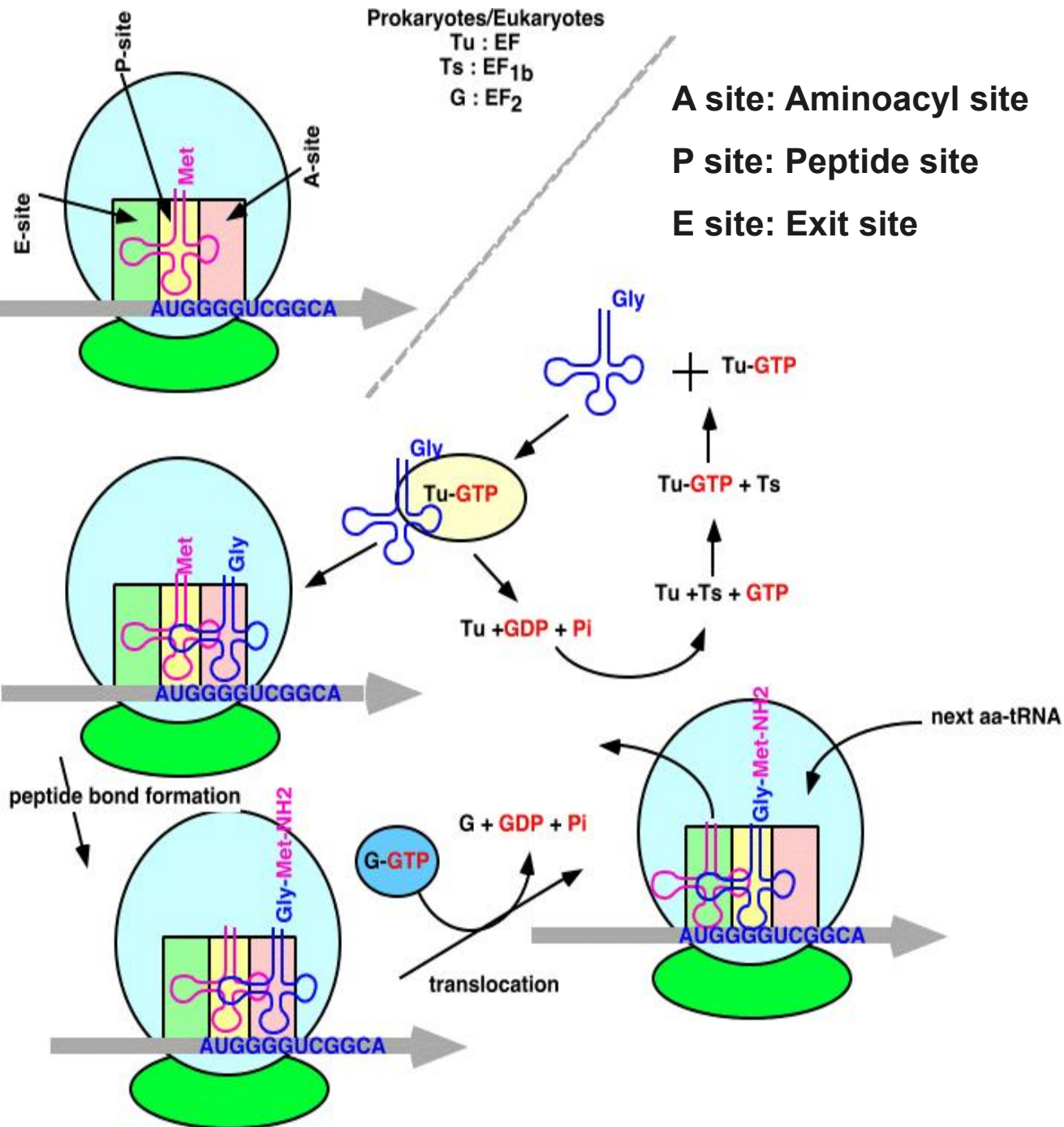
The Code is Universal: not always true

- As more gene and protein sequence information has been obtained, it has become clear that the genetic codon is not strictly universal.
- In certain eukaryotic nuclei and mitochondria codons that cause termination in the standard genetic code can code for amino acids such as tryptophan and glutamine.
- In several mitochondrial genomes, the sense of codon is changed from one amino acid to another.

Table 18.1 Deviations from the “Universal” Genetic Code

Source	Codon	Usual Meaning	New Meaning
Fruit fly mitochondria	UGA	Stop	Tryptophan
	AGA & AGG	Arginine	Serine
	AUA	Isoleucine	Methionine
Mammalian mitochondria	AGA & AGG	Arginine	Stop
	AUA	Isoleucine	Methionine
	UGA	Stop	Tryptophan
Yeast mitochondria	CUN*	Leucine	Threonine
	AUA	Isoleucine	Methionine
	UGA	Stop	Tryptophan
Higher plant mitochondria	UGA	Stop	Tryptophan
	CGG	Arginine	Tryptophan
<i>Candida albicans</i> nuclei	CTG	Leucine	Serine
Protozoa nuclei	UAA & UAG	Stop	Glutamine
<i>Mycoplasma</i>	UGA	Stop	Tryptophan

*N = Any base



EF-Tu:

EF-Ts:

EF-G:

Elongation:

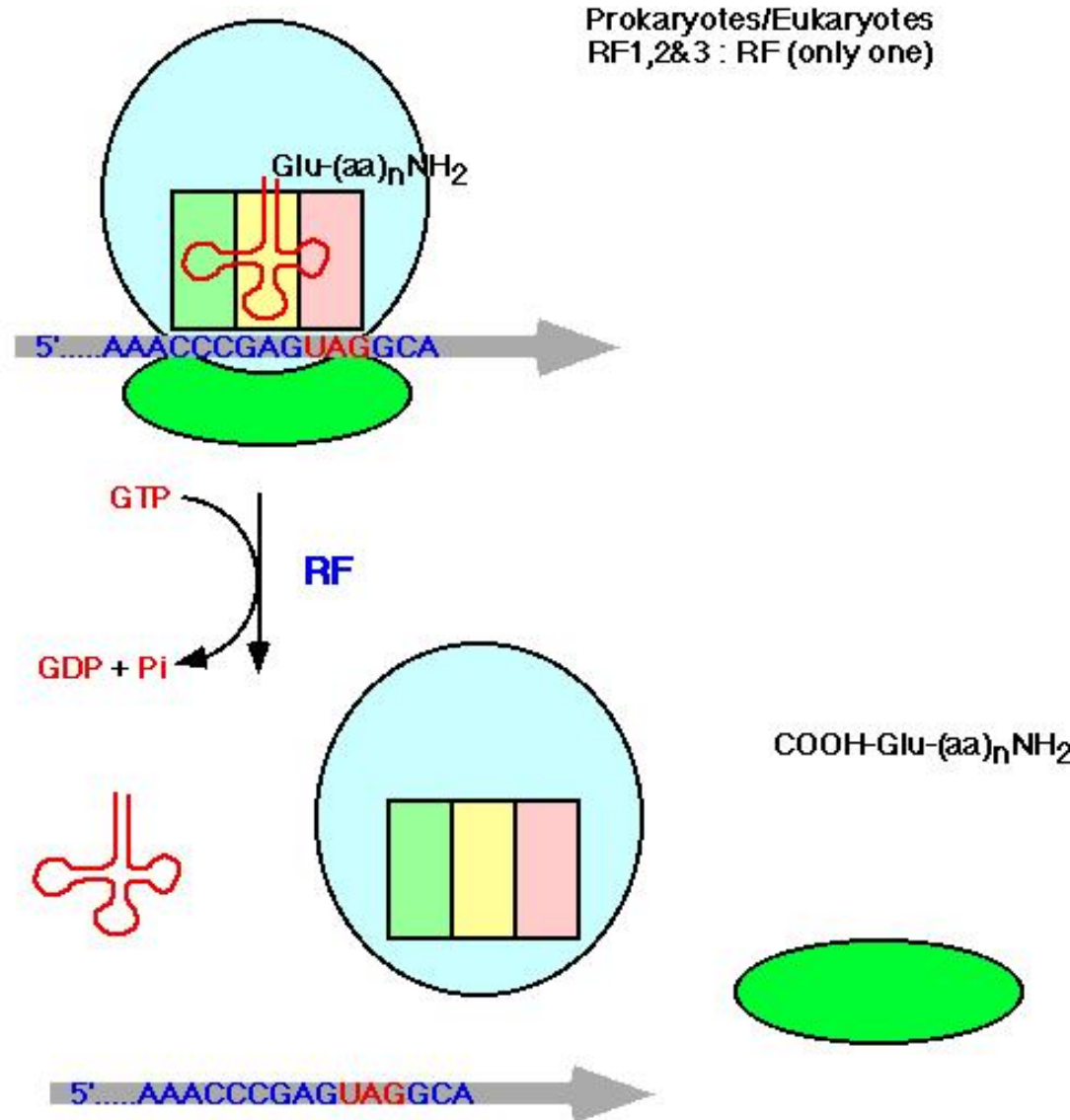
1. Entry (binding) of the next aminoacyl-tRNA to the A (aminoacyl) site
2. Formation of the Peptide Bond
3. Translocation

The role of three types of EFs (elongation factors)

- EF-Tu: EF-Tu –GTP escort aminoacyl-tRNAs to the ribosome
- EF-Ts: regenerates EF-Tu-GTP
- EF-G: help translocation

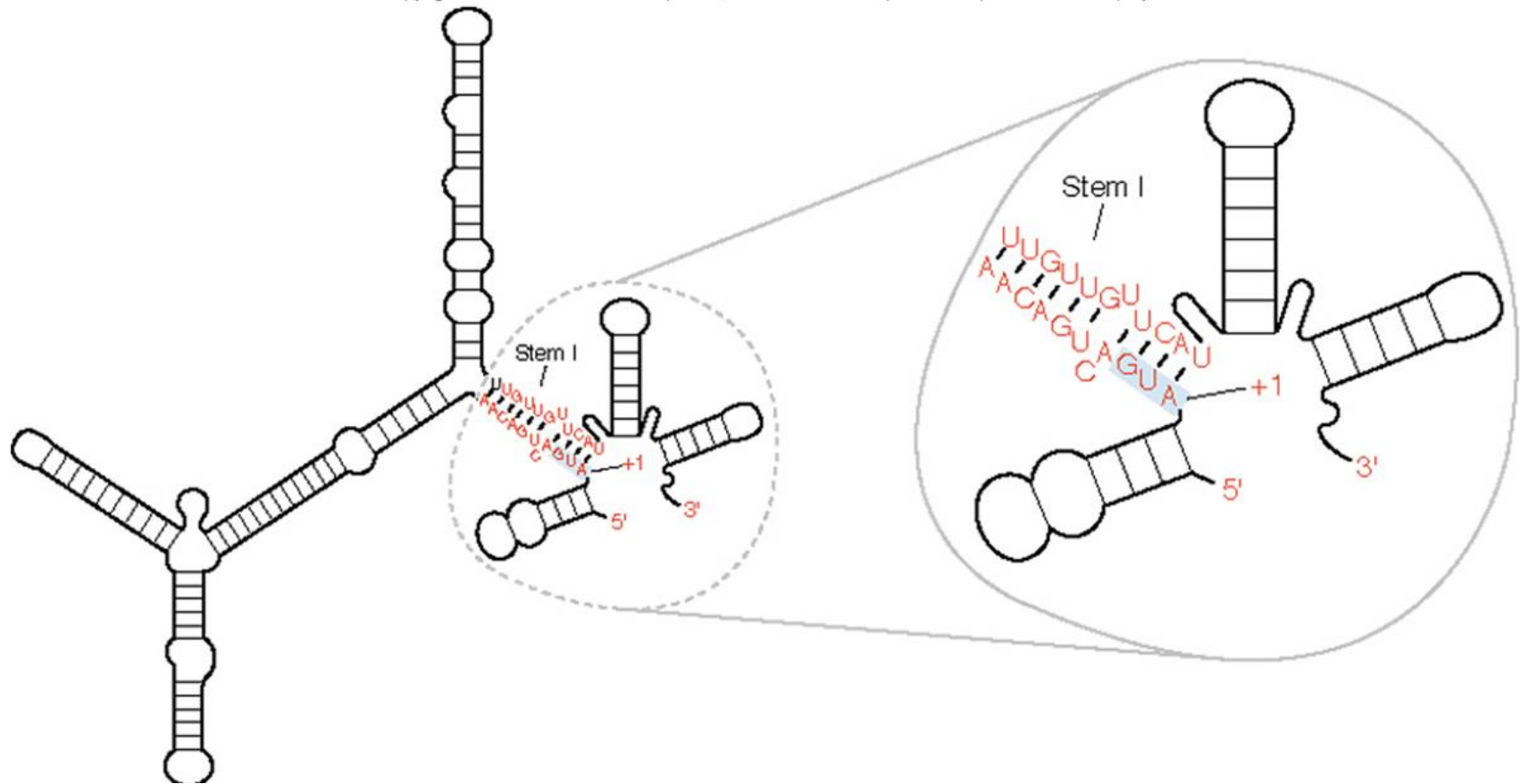
Termination of translation

- The “A” site of the ribosome over one of the stop codons.
- There will be no corresponding aminoacyl-tRNA
- Release factors (RFs) enter the A site
- Ribosome is released from the mRNA



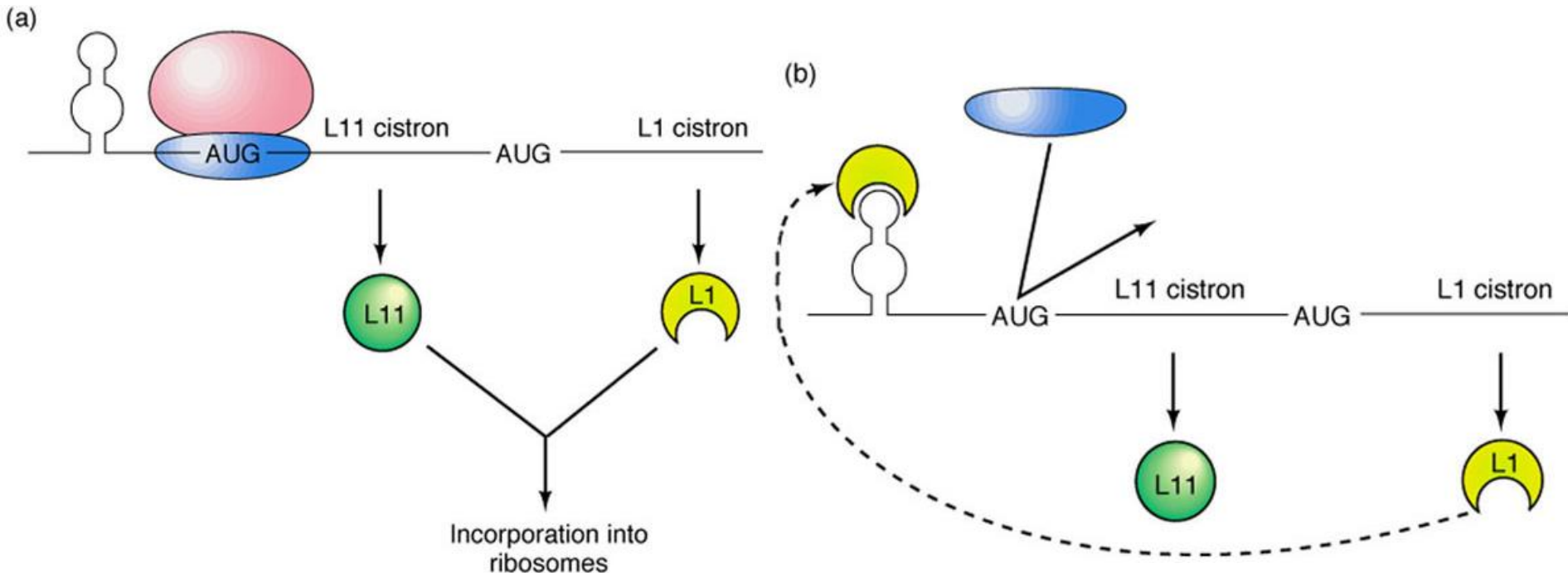
Translational control

- **Prokaryotic translational control: in initiation stage**
Prokaryotic mRNAs are very short-lived, so control of translation is not common in these organisms. However, some translational control does occur.
 - mRNA secondary structure can govern translation initiation (**σ^{32}** accumulating in less than a minute after heat shock.)
 - Controlled by feed-back repression



- Under normal temperature 37°C, there is the base pairing between the initiation codon (boxed) and a region near the 3'-end of the mRNA, forming "stem I, which prevent the translation of σ^{32} mRNA.
- When heat shock (42 °C), σ^{32} begins accumulating in less than a minute, which is too little time for regulation on transcriptional level.
- When the temperature rises, the base pairs causing this secondary structure melt, unmasking the initiation codon.

Feed-back repression



Polycistronic mRNAs encoding the *E. coli* ribosomal protein (L11, L1) genes
(a). When too much L11 and L1 proteins are made. The excess L1 finds no 23S rRNA for which to bind, so it binds to the stem loop near the initiation codon to block the translation (b).

Eukaryotic translational control

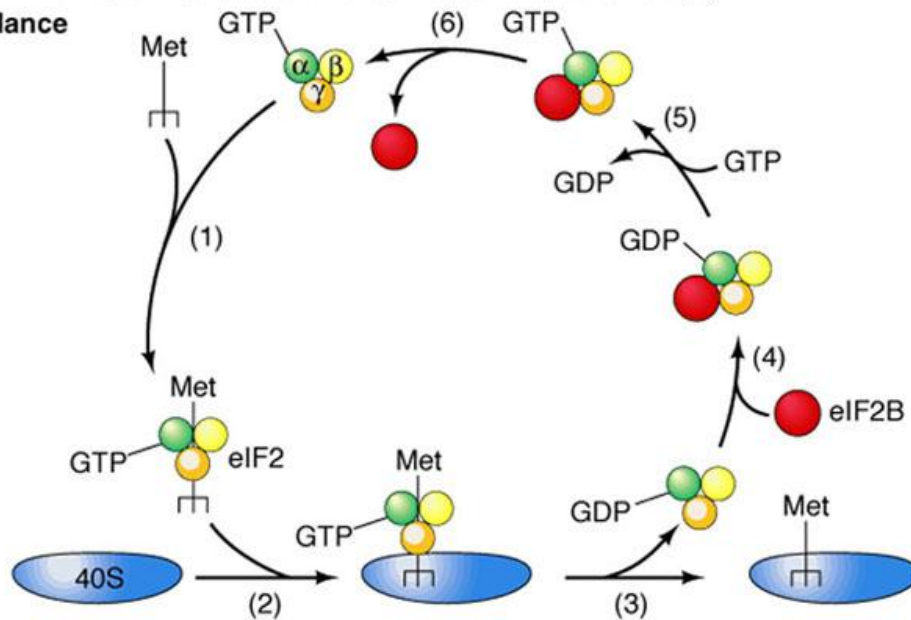
Eukaryotic mRNA lifetimes are relatively long, so there is more opportunity for translation control than in prokaryotes.

Eukaryotic translational control

Diverse **mechanisms** of eukaryotic regulation include, e.g.:

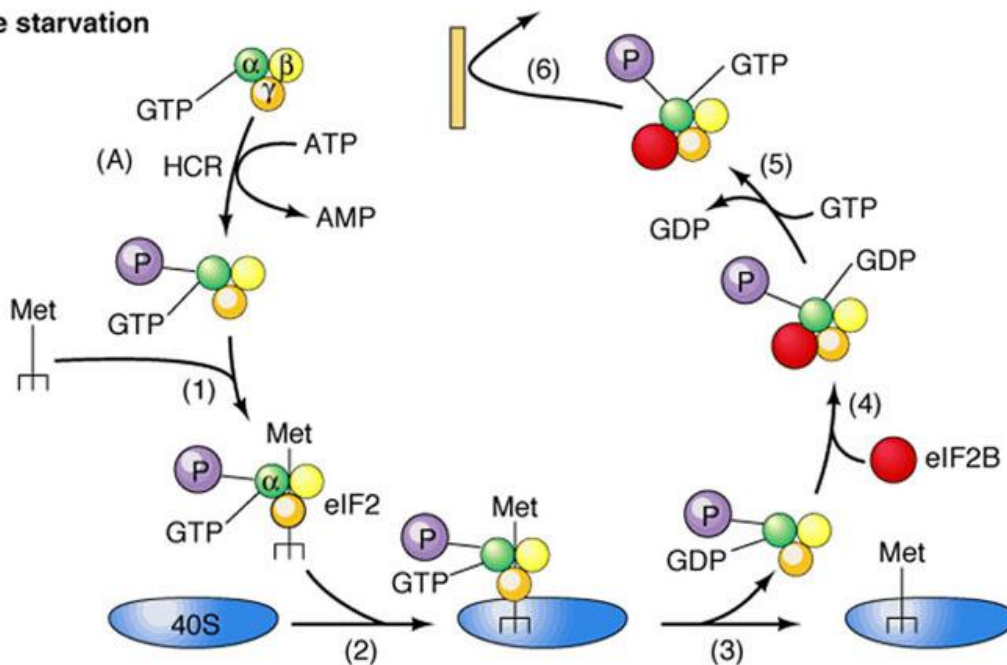
- Signal-activated **phosphorylation** or **dephosphorylation** of initiation and elongation factors
- **Inhibition** of translation by small **single-stranded RNA** molecules, **miRNAs**, that bind via base-pairing to 3' un-translated regions of mRNA.

(a) Heme abundance



The α -subunit of eIF2 is a favorite target for translation control.

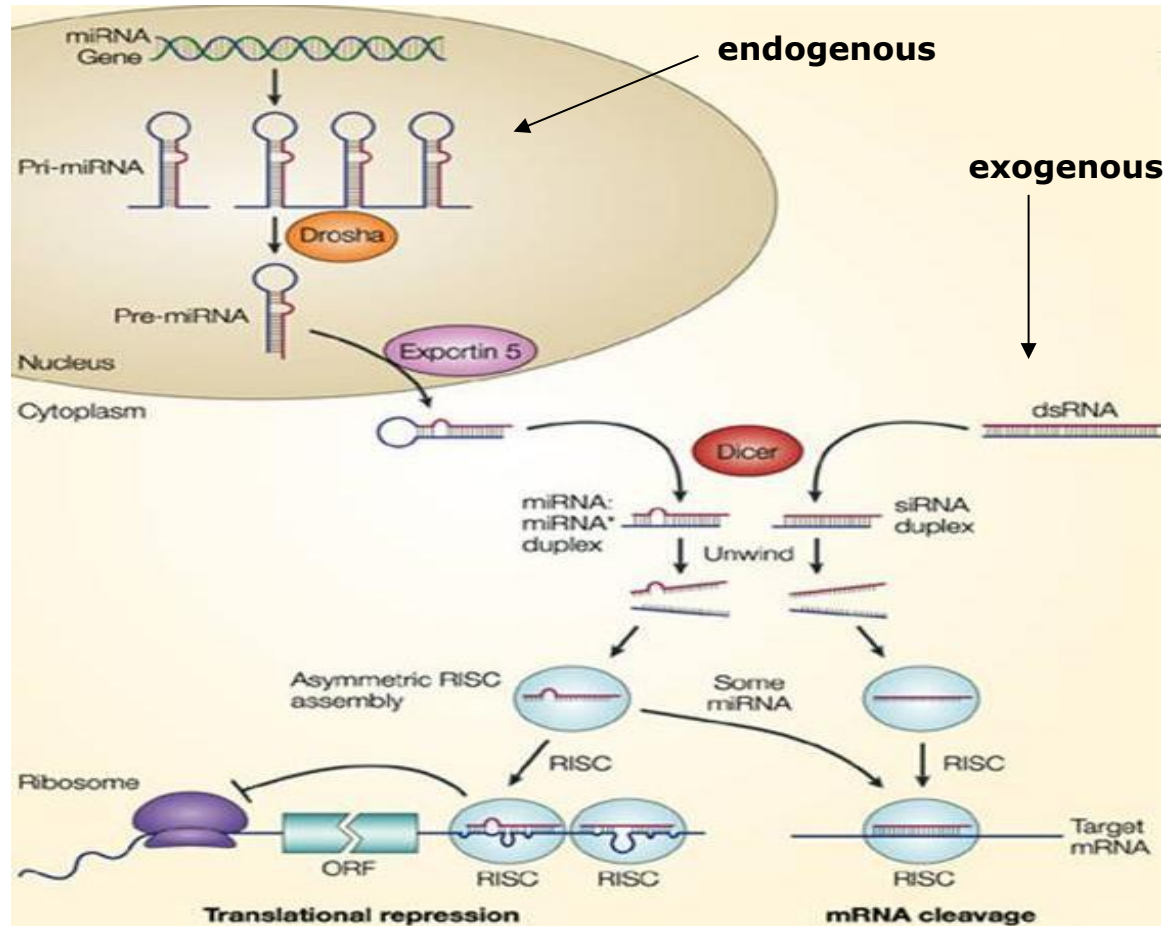
(b) Heme starvation



In heme-starved reticulocytes (b), HCR is activated, so it can phosphorylate eIF2 α and inhibit initiation. Step 6 (in b) is blocked because the high affinity of eIF2B for the phosphorylated eIF2 α prevents its dissociation.

Two kinds of control mechanism of miRNA

mRNA cleavage vs. Translational Repression



Post-translational Modifications

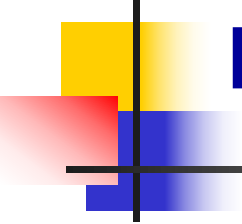
- **Post-translational Modifications**: Some of the standard amino acids can be modified after incorporation into peptides or proteins
- **Cleavage** : to remove signal peptides, to release mature fragments from polyprotein (ubiquitin), to remove internal peptides as well as trimming both N-and C-termini
- **Chemical modifications**: take place on the N-and C-termini ,as well as on all but six (Ala, Gly, Ile, Leu, Met and Val) of the amino acid side chains. Include acetylation (乙酰基化), hydroxylation (羟化) , phosphorylation (磷酸化) , methylation, glycosylation (糖基化)
- **Disulfide bond**

Post-translational Modification

- **Modification occurs at specific residues: not random**
- **Modifications are enzyme catalyzed**
- **The enzymes that catalyze the modification recognize a specific amino acid sequence in the protein that is modified**

Example: Casein kinase II phosphorylation site: [ST]x(2)[DE]

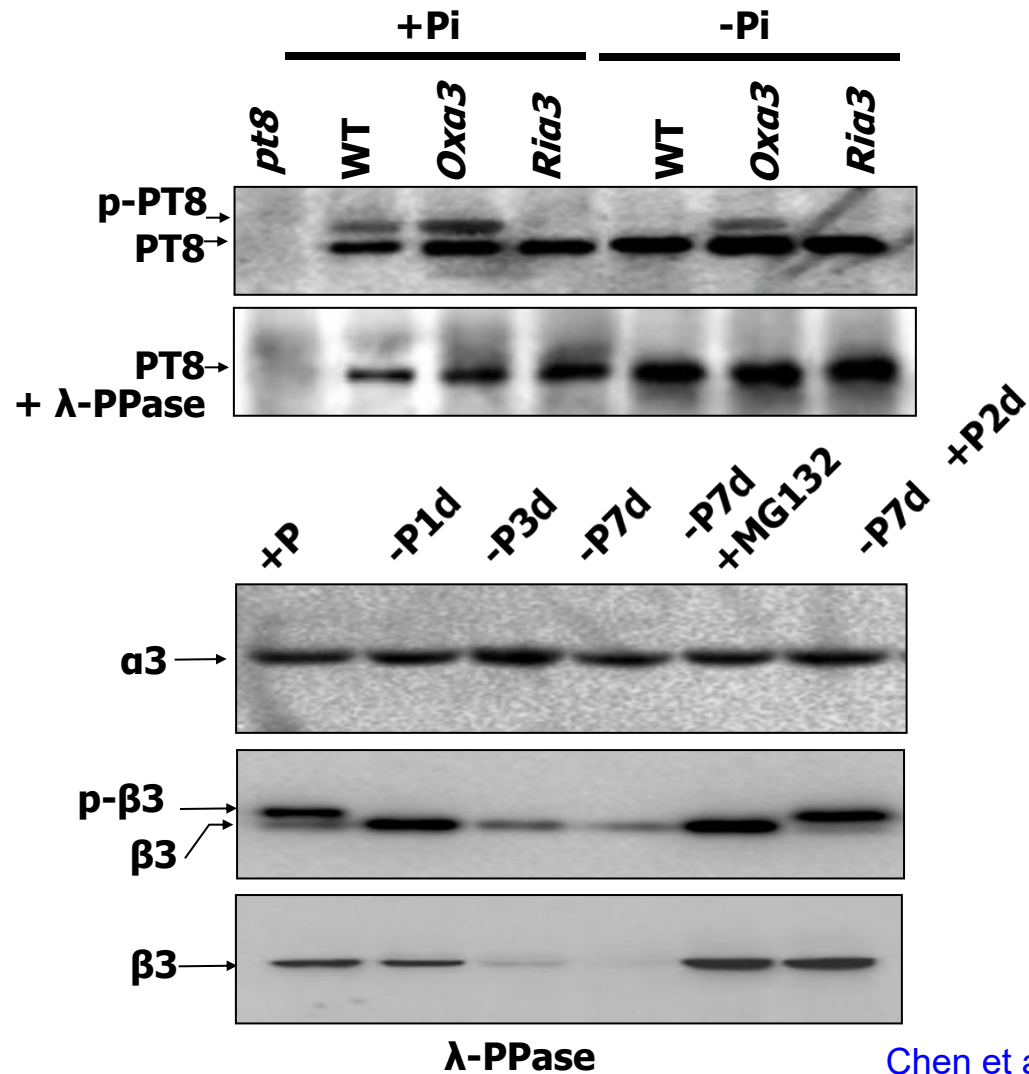
protein kinase C phosphorylation site: [ST]x[RK]



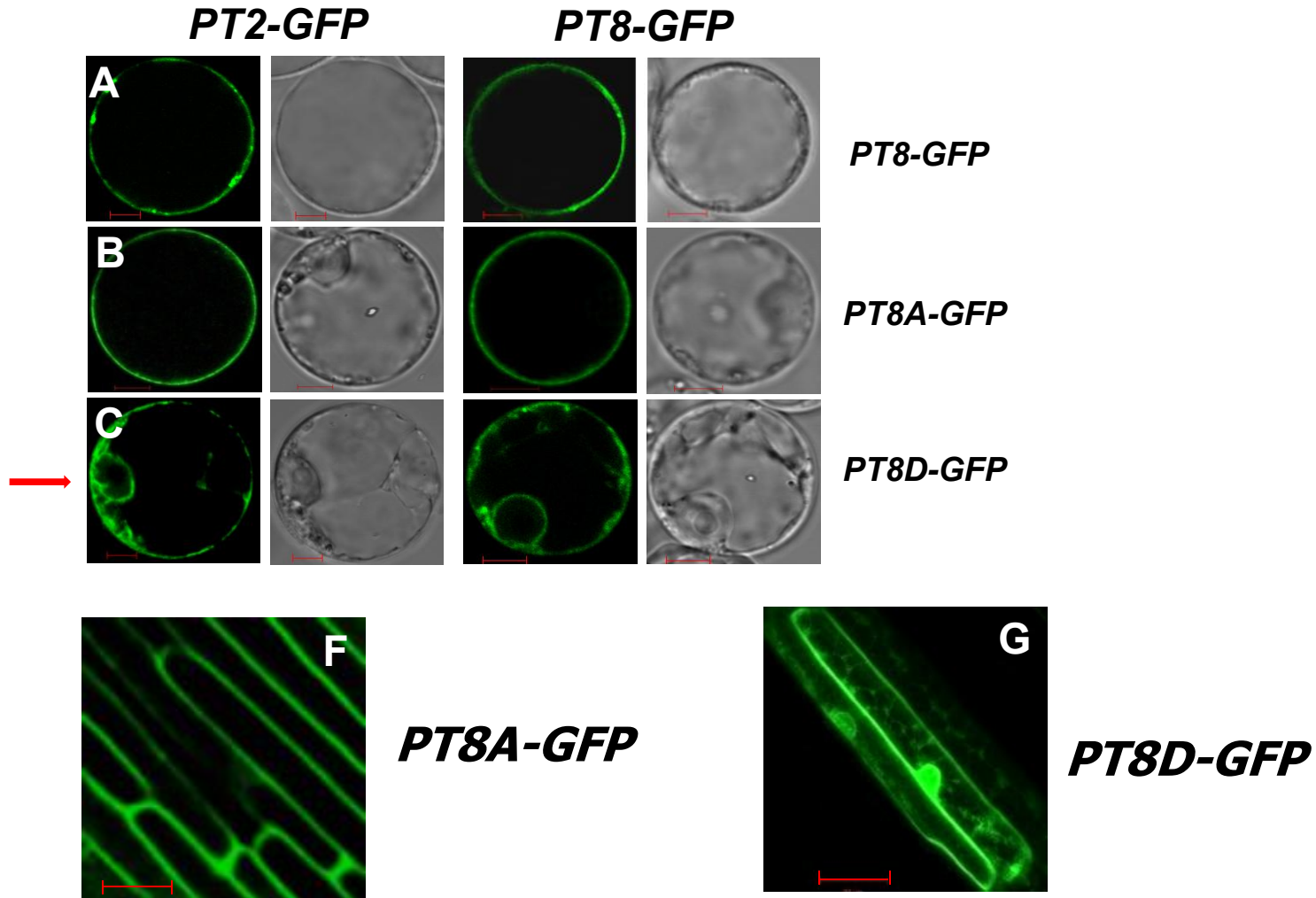
Post-translational Modification

- Modified residues have important roles in protein function.
- OH on hydroxyLys & hydroxyPro in collagen (胶原蛋白) are critical in H-bond stabilization of triple helix
- Sugars attached to OH of Ser, Thr are critical in recognition of receptors on cell surface, transport to organelles, etc.

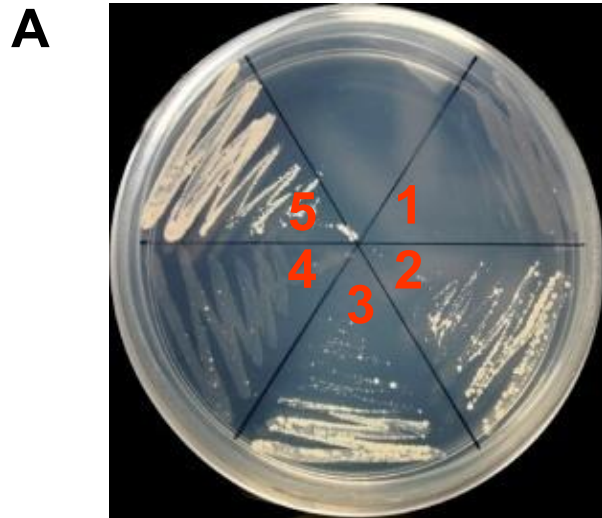
Example: OsCK2 磷酸化 PT



磷酸化影响 **PT** 从内质网输出到质膜



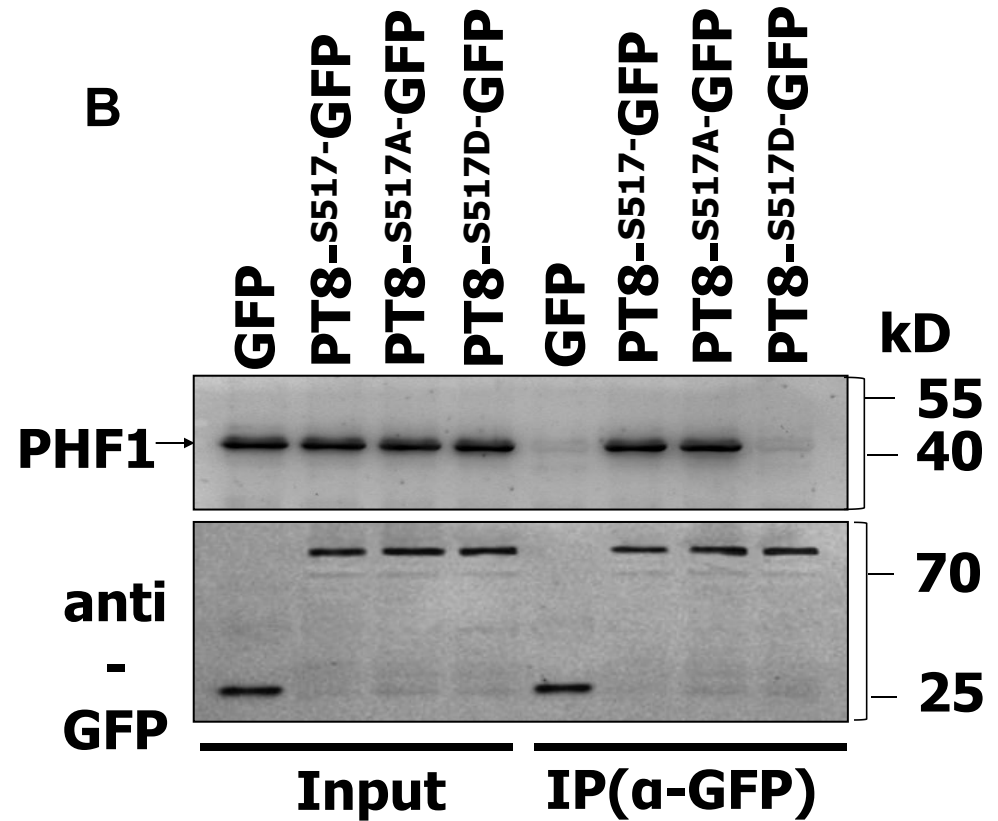
517位置磷酸化抑制PT与OsPHF1互作



1, Negative

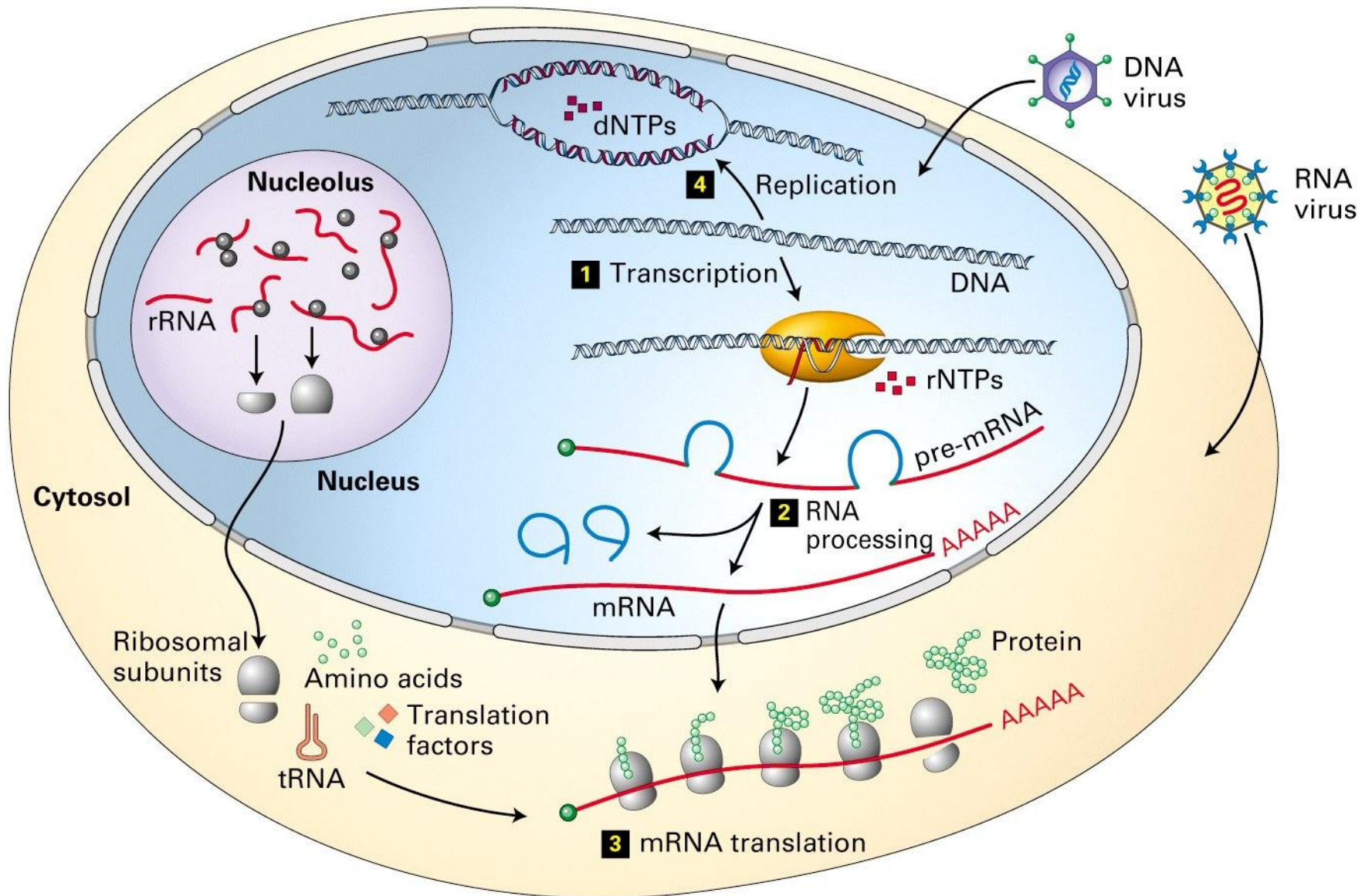
2, PT8^{S517}; 3, PT8^{S517A}

4, PT8^{S517D}; 5, Positive



Summary

- Components of translation: ribosome, mRNA, tRNA, aa
- Initiation: IFs, AUG, consensus sequence
- Elongation: EFs
 - Direction: N-----C
 - Codon: base-pair rules
- Termination: UAA, UAG, UGA
- Translational control
- Post translational modification



Protein Synthesis



Play



Pause

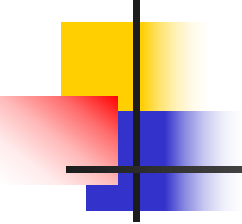


Audio



Text

In prokaryotic cells, translation is initiated by formation of an initiation complex consisting of the 30S ribosomal subunit, formyl-methionyl (f-Met) tRNA, and messenger RNA.



Student's presentation (today)

- Paper: Fu, et al. The TOR–EIN2 axis mediates nuclear signalling to modulate plant growth. Nature 591:288-292 (2021)
- Presenter: Group 11